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Flexible Automation in Air Traffic Control Through Adaptation of Human-Automation Collaboration

Magnus Nylin



LINKÖPING UNIVERSITY

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SAMMANFATTNING

Ökad automatisering av tekniska system pågår inom många områden. För att den ökade automationsgraden ska ge önskad nytta måste också samarbetet mellan människa och automation utvecklas, annars riskerar man att få problem med exempelvis automationsöverraskningar och minskad situationsmedvetenhet hos användarna. I säkerhetskritiska domäner som flygtrafikledning kan det i värsta fall leda till direkt farliga situationer. Trots att utmaningarna och riskerna med automation är välkända återkommer problemen ofta i nya system. Det tyder på att frågorna inte hanteras tillräckligt i utformningen av ny automation och att det behövs bättre metoder och idéer för att identifiera och motverka problemen. I avhandlingen utforskar jag hur konceptet följsam automation kan öka förståelsen för hur ny automation ska designas för att skapa ett bättre samarbete mellan människa och automation. Den grundläggande idén med följsam automation är att ge automationen kunskap om människans situation och kombinera den med kunskap om de processer som människan och automationen tillsammans arbetar med. Konceptet vidgar synen på automation från att enbart ersätta människan med teknik till ett perspektiv där den totala prestationen människa-automation blir större än summan av de enskilda delarna och där människan fortfarande spelar en viktig roll. Det här skulle kunna tillämpas inom många olika områden, men fokus för avhandlingen är flygtrafikledningsdomänen, även om den till viss del också berör sjötrafikinformationstjänst och tågtrafikledning.

Utgångspunkten i avhandlingen är de redan kända utmaningarna med ökad automation och från det har problematiken brutits ner för att hitta förslag på lösningar. Ett avgörande moment i samarbetet mellan människa och automation är när automationen initierar kommunikation med människan. Den insikten har resulterat i två centrala koncept som presenteras i avhandlingen. Det första konceptet är designansatsen *Reduced Autonomy Workspace* (RAW) som beskriver hur anpassning av kommunikation mellan automation och människa kan utformas för situationer när automationen måste konsultera människan. De temporala aspekterna är mycket viktiga och designansatsen inkluderar också hur automationen ska transformera informationen till ett format som kräver så lite kognitiv ansträngning som möjligt för människan.

En glyfbaserad visualisering för användning i framtida högautomatiserade system baserad på designansatsen RAW togs fram och utvärderades i ett antal workshops med operativa domänexperter från flygtrafikledning, sjötrafikinformationstjänst och tågtrafikledning. Det andra konceptet är en ny typ av visualiseringskoncept för uppmärksamhetsstöd, *Soft Visual Cues (SVC)*, eller mjuka visuella påminnelser. Mjuka visuella påminnelser kompletterar andra typer av uppmärksamhetsstöd som aktivt drar uppmärksamheten till sig, vilket kan vara ett problem om användaren är upptagen med något viktigare som automationen på grund av en bristande helhetsbild inte känner till. Olika designalternativ utvärderades i workshopform tillsammans med flygledare och den föredragna designen testades sedan i realtidssimuleringar med andra flygledare. Utöver de framarbetade koncepten har också studier genomförts för att skapa en djupare förståelse för de ingående kontrollprocesserna, hur de påverkar varandra och hur domänspecifika förutsättningar avspeglas i dem.

För att förstå de processer som utgör samarbetet mellan människa och automation så har flera av studierna involverat domänexperter, till exempel flygledare. Insamlingen av data genom workshops och realtidssimuleringar har varit en utmaning utöver det vanliga då en stor del av arbetet utfördes under Covid-19-pandemin, men trots detta har det mesta gått att genomföra som planerat. Resultaten har bidragit till en utökad kunskap om design av interaktion mellan automation och människa, vikten av temporala aspekter, samt hur det kan visualiseras. Designansatsen RAW och glyferna bidrar med kunskap främst riktad mot framtida system, medan mjuka visuella påminnelser är en lösning som skulle kunna implementeras i närtid. Problemen och de föreslagna lösningsansatserna i avhandlingen är huvudsakligen applicerade på flygtrafikledning. Den grundläggande problematiken runt samarbete människa-automation i kontroll av realtidsprocesser är dock densamma inom många områden. De presenterade idéerna och koncepten bör därför kunna tillämpas även inom andra domäner och ge goda förutsättningar för att med människan som utgångspunkt designa framtidens följsamma automation.

ABSTRACT

Many domains currently experience an increase in level of automation of their technical systems. However, the increased level of automation must be accompanied with the development of well-functioning human-automation collaboration to avoid undesirable phenomena such as automation surprises or reduced situational awareness. In safety critical domains like air traffic control, this may be more than an inconvenience or loss of efficiency: in the worst case, it can jeopardize flight safety. Although these issues and risks related to increased automation are well-known, they continue to appear in new systems. This indicates that they are not sufficiently addressed in the design of new automation, and that better methods and ideas for this must be developed. This dissertation explores how the concept of flexible automation can be used to increase the understanding of how to design for better human-automation collaboration. The fundamental idea of the flexible automation concept is to provide the automation with knowledge about the human's situation and let the automation combine that with its understanding of the overall situation, including mutual work- and control processes. At a conceptual level, this also widens the view on automation as merely a way to replace human labor, to a view on automation where it enhances the joint performance of the human-automation team. It also emphasizes the human as an essential piece of a successful automation. Though being important and applicable to many areas and domains, this dissertation is mainly focused on the air traffic control domain, but also includes elements from the domains of vessel traffic service and train traffic management.

The approach for the dissertation has been to use the known problems and issues with automation as a starting point, and then decompose that complex of problems and explore them to find possible solutions. Initiation of communication between automation and human, how and when it is performed, were identified as one critical point in the human-automation collaboration. This insight formed the basis for the two main concepts presented in this dissertation. The first concept is the *Reduced Autonomy Workspace* (RAW), a design approach that describes the process of when a highly autonomous automation needs to consult the human working together with the automation. Focus is on the temporal aspects, but it also includes transformation of the

information content to reduce the cognitive effort on the human receiving the information. A visualization for future automation using glyphs based on the same principles was developed to support the RAW design approach. The glyphs were evaluated in workshops with operators from air traffic control, vessel traffic service, and train traffic management, respectively. The second main concept is a new type of attention support, *Soft Visual Cues* (SVC). The SVC complements other types of attention support which may, due to the automation's lack of knowledge of the overall situation, draw the operator's attention away from something potentially more important. Different designs were evaluated in operator workshops and tested by other air traffic controllers in a series of real-time simulations. In addition to these concepts, studies have also been performed to explore and gain a deeper knowledge of how the involved control processes can be understood, the effect of entangled control processes, and how domain specific prerequisites are reflected in the control processes.

The studies have involved subject matter experts, such as air traffic controllers, to understand the processes that underlies human-automation teamwork. However, data collection through workshops and simulations has been challenging because of the pandemic, even though most activities were possible to perform. The results have contributed to a better understanding of the design for human-automation collaboration, the importance of temporal aspects in this, and how it can be visualized. While the RAW and the glyph visualizations aims at future automations, the SVC is a concept that could be implemented in a near future. The problems and proposed solutions in this dissertation are mainly focused on applications in air traffic control. However, the problems and issues addressed here are similar in many other areas focused on control of real-time processes. Therefore, the presented ideas and concepts should be of value for those as well, and the results are a good starting point for the development of tomorrow's flexible automation, where the human is still a major asset.

ACKNOWLEDGMENTS

My PhD journey started in the autumn of 2014. During a local LFV Research and Development department meeting, the possibility to have industrial PhD students at LFV was mentioned in passing, and I recognized this as an opportunity too good to be missed. Some time and a M.Sc. later, I started my PhD studies in August 2018, of which this dissertation is the result. However, I would never have got here without the support from many others. My supervisors, Jonas Lundberg (main supervisor), Jimmy Johansson Westberg, and Kostiantyn Kucher (co-supervisors), have played a most essential part in guiding me through five intense years, by sharing their academical expertise, by supporting me, by challenging me, and by cheering me up when so needed. For managing the F AUTO project and developing much of the technology I have used in my studies, Magnus Bång and his excellent team of developers have been a valuable support. To even get started, LFVs one and only Billy Josefsson has played an important role in establishing the cooperation with Linköping University that eventually led to the possibility for me to start my PhD, to my knowledge being the first ever industrial PhD student at LFV. This also includes everyone else at the LFV Research and Innovation department who has believed in me and made this possible. Of these, I want to mention my fellow PhD student and colleague since many years, Jens Nilsson, with whom I have had endless discussions about everything. Finally, the way towards a PhD requires commitment and time. It also includes periods of doubt and moments when the final goal seems very far away. During these times, as well in all good times, my wife Christina has, except for being my life companion and loved one, been my most valuable supporter in all this.

To all above and those not explicitly mentioned but who have been a part of this, thanks for your support and that you have made this possible for me!

INCLUDED PAPERS

Paper I
Attention Support With Soft Visual Cues (Nylin, M., Lundberg, J., Johansson, J.), *IV 2020: 24th International Conference on Information Visualisation*, IEEE, 2020, DOI: 10.1109/IV51561.2020.00035

Paper II
Reduced Autonomy Workspace (RAW) — An Interaction Design Approach for Human-Automation Cooperation (Nylin, M., Johansson Westberg, J., Lundberg, J.), *Cognition Technology and Work*, Springer, 2022, DOI: 10.1007/s10111-022-00695-2

Paper III
Reminding Without Disturbing: A Soft Visual Cues Attention Guidance Design for an Air Traffic Control Conflict Detection Visualization (Nylin, M., Söderholm, G., Lundberg, J., Bång M.) [1]

Paper IV
Glyph Design for Communication Initiation in Real-Time Human-Automation Collaboration (Nylin, M., Lundberg, J., Bång M., Kucher, K.) [2]

Paper V
Modelling of Operator Control Work Across Traffic Management Domains: Implications for Interaction Design (Lundberg, J., Nylin, M., Praetorius, G., Arweström Jansson, A., Bång, M.) [1]

[1]Submitted October 2023
[2]Submitted September 2023

ADDITIONAL PAPERS

Paper VI
Adaptive Automation Through Interactive Visualization (Nylin, M.), *Proceedings of the 31st European Conference on Cognitive Ergonomics: Design for Cognition (ECCE 2019)*, ACM, 2019, p. 26–27, DOI: 10.1145/3335082.3335123
NOTE: Paper for doctoral consortium.

Paper VII
Untangling Complexity in Control Process Interactions (Nylin, M., Vrotsou, K., Bång, M., Lundberg, J.), *Position paper for the Emergent Interaction Workshop, CHI2021*, 2021, Workshop, https://emergentinteraction.github.io/assets/untangling_complexity.pdf

Paper VIII
ATCO Strategies in Tactical TBO: A Simulator Study (Nylin, M., Nilsson, J., Lundberg, J., Bång M.), *ICAS 2022: 33rd Concress of the International Council of the Aeronautical Sciences*, 2022

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1 Introduction

Automation and digitalization are changing the way we work. With the development of increasingly sophisticated technologies, such as artificial intelligence, automation is increasing in many areas in society. The goal is often to help us to perform better, smarter, faster, or cheaper, or to achieve things that have never been possible before. However, human-automation collaboration needs to develop accordingly to make it possible to take advantage of the technological development and to avoid classical automation problems such as automation surprises [80], loss of situation awareness (see, e.g., [19, 37, 46]), or the automation ironies articulated by Bainbridge [2] in 1983. The latter includes phenomena like the need for higher skills than before to master situations when the automation is failing, or that the automation adds to the total workload instead of reducing it because the operator is still making some decisions *and* monitors the automation. Furthermore, increased automation may introduce new risks that we do not know about yet. Those must be identified and mitigated, existing theory must be questioned, developed, and complemented to cover the new human-automation collaboration landscape. If not so, possible gains from introducing automation may be counteracted by badly designed human-automation collaboration, or in the worst case, creating hazardous situations in safety critical domains, such

as Air Traffic Control (ATC). Much research has been conducted within the field of automation and human-automation collaboration, however, many of the identified challenges remain, for example, how to avoid the automation ironies [4, 21, 88].

One approach to enhance human-automation collaboration is to let the automation adapt to the human in a flexible way. The study of, and design for, flexible automation is the main topic for this dissertation. The research has been performed within the setting of the flexible automation (F AUTO) project. This is a domain transversal project which includes research on human-automation collaboration in control rooms environments within ATC, Vessel Traffic Service (VTS), and Train Traffic Management (TTM). Even though the research was carried out within these specific domains, the knowledge produced is intended to be generalizable to other domains with similar characteristics. However, my focus for the dissertation, as well as my main responsibility in the F AUTO project, has been ATC.

The notion of flexible automation is central to my work and for the F AUTO project. By using the term *flexible automation*, we wanted to emphasize that the automation shall not only adapt to the human; it shall do so in a smooth and sophisticated way based on an understanding of the human's situation, thus increasing the total performance of the system together with the human. This is an extension from a more mechanical view on automation adaptation, where the adaptation only follows a logical pattern of ifs and thens. The logical patterns may cover some adaptation needs, but it will be impossible to cover complex situations where all states and events cannot be predicted. To achieve the full flexibility, the automation must also possess knowledge of, and consider, the process(es) which it controls together with the human, with an emphasis on *together*. As long as the automation is not completely autonomous, there will always be an interaction with a human at some stage. The idea of the flexible automation is based on the belief that a well-functioning automation is easier, and maybe even only possible, to achieve if it is designed with the goal to assist humans rather than to completely replace them, even if the result may involve less human effort per given output. Early in my work, I identified the initiation of communication from the automation to the human as a critical moment. Hence, much of my research on flexible automation has revolved around that information exchange initiation process, not least for understanding the temporal aspects of the communication.

1.1 Motivation

So, if the problems and issues related to automation are well known and have been studied for at least the past forty years, *why do they still occur*? Why do we still produce technically ingenious solutions that nevertheless end up confusing and frustrating the human users? In the beginning of my work in the F AUTO project, the focus was on adaptive automation as such, and how it could be designed to also be flexible in line with the F AUTO ideas. As pointed out by Baxter et al. [4] and Strauch [88] (in 2012 and 2018 respectively), the fundamental problems with automation articulated by the automation ironies seemed not to have been solved, and recently Endsley came to the same conclusion; the issues still remain [21]. In addition, Endsley also observed that the introduction of artificial intelligence may add new ironies. From an overview of research on adaptive automation early in my work, it became apparent that adaptive automation alone did not seem to solve the problem, even though it could be part of the solution. There were several examples of adaptive automation that did not fully deliver the intended results. For example: consider that a system is able to identify that the human operator is starting to get overloaded and steps in to help by taking over certain tasks. If well designed, that may function well, but what happens next when the situation has calmed down — how should the human regain control from the automation? As pointed out by, for example, Salomon and Boudreaux [76], this might be just as important as identifying when to initiate the help. When the automation has stepped in, temporarily increasing the level of automation, it is likely that the human is not aware of the overall status of the system and what the automation has done since it took over. Thus, the automation would have to explain this bit by bit to the operator. However, unless the processes can be stopped, or be left unattended for some time, this has to be done while both the human and the automation are still performing tasks. Alternatively, the operator must keep track of what the automation is doing throughout the process to be able to regain control, but the cognitive resources freed up by the support of the automation are then spent on monitoring the automation instead (which is one of the ironies described by Bainbridge [2]). Another insight gained was that there were numerous notions of levels of automation (see, e.g., [94]) which describe how tasks should be divided between automation and human, but there was a lack of ideas for, or description of, how humans and automation

should communicate around the tasks to be performed, both between and within different levels of automation.

The changed behavior of an automation, or the level of automation, may be the source of certain automation issues, but there are other problems related to the exchange of information between the human and the automation. Even if the level of automation is kept constant, the automation and the human must exchange information about the ongoing control processes in which both are involved to be able to cooperate. Humans working together constantly interact to assess why, what, and when to communicate. We understand the other person's situation and adapt our communication accordingly, and if we are trained to perform specific tasks together in a team, we can develop these skills further. Most technical, automated, systems do not have this ability. This lack of understanding and lack of ability to interpret the person the automation works together with may lead to inconveniently initiated communication. It might not contain the information needed, it might contain information certainly *not* needed, it might be initiated when the human is not ready to perceive it, or when there is no time to act on what is communicated. An everyday example of this is pop-up tooltips provided by office desktop applications, or surveys in on-line applications. They rarely pop up when help on a certain subject is needed, and they certainly do not consider whether the user is in a state to take in the information. In an office environment, this behavior is most likely just annoying. In a real-time, safety critical environment, the systems are hopefully not crammed with pop-up tooltips and the user interfaces are more worked through. Yet, the challenges of initiating communication timely and with a relevant content are the same. Like the issues related to transferring control between automations and humans, this is due to a lack of ability to understand the humans' situation, and there is no immediate solution in sight.

Unless we give up and establish that we have to live with these problems and issues, we must look for another way to approach the problem, even if there is no silver bullet solution. The idea emerged that it must be something in the process of designing automation that is missing. Could it be that the problems are addressed too late in the design process? When automated systems are designed, the starting point is often the goal of replacing human labor and increase efficiency by automating certain tasks. If we instead flip the perspectives and start with the already known problems and issues and think about how automation can be designed to avoid them al-

ready from the beginning, maybe it should be possible to understand where it goes wrong and to formulate a possible way forward? Humans possess incredible abilities, and these shall not be forgotten or lost when automating. The F AUTO approach, both within the project and in my own research, is an attempt to shift focus from aiming at gradually replacing the human, to a view on automation where the goal is to increase the total output from the human-automation collaboration by supporting the humans to perform better together with the automation. In the long run, this may nevertheless lead to that fewer persons are needed to perform a certain task, but hopefully they will be happier and better motivated human operators that work well together with, and embrace, the automation.

1.2 Project and Participants

All my work included in this dissertation was performed within the frames of the F AUTO project and can be seen as sub-parts of the project, in which several stakeholders were involved. The project had a cross-domain approach and consisted of one branch per domain, where each branch had a research partner from academia and one service provider organization from the respective domain, as demonstrated in Fig. 1.1. As an Air Navigation Services of Sweden (LFV) PhD student, I had affiliations to both LFV and Linköping University and had the main responsibility for the ATC branch of the project. A researcher from Linnaeus University (later affiliated with Swedish Road and Transportation Research Institute (VTI)) together with the Swedish Maritime Administration was responsible for the VTS branch, and a researcher from Uppsala University together with the Swedish Transport Administration was responsible for the TTM branch. In addition, researchers and research engineers from Linköping University were responsible for a fourth, transport domain transversal technology branch that performed technical developments to support the project studies, as well as research on future technology, mainly related to operator sensor technology and AI-applications for interpreting the sensor data.

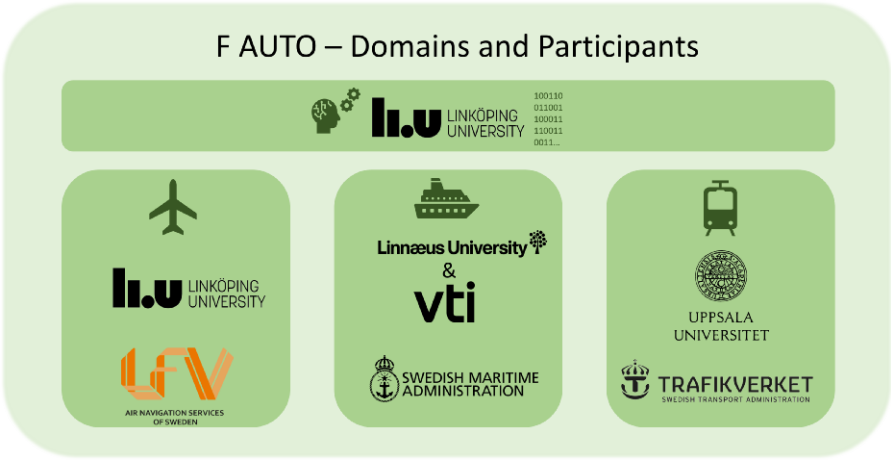


Figure 1.1: The F AUTO project's different partners per domain. In addition, there was a branch for technology development and research which supported all domains.

1.3 Aim and Research Questions

The aim of this dissertation is to explore the concept of flexible automation in control room environments, with a focus on how to design flexible automation that avoids classical automation traps and failures.

My work has an applied perspective of designing well-functioning automation, but to achieve that, there must also be an understanding of how collaboration between humans and automation in these applied situations could work. This understanding requires research on the underlying mechanisms, which has been important for my dissertation. Examples of this are new insights about temporal aspects in human-automation collaboration and the role of interactive visualization to support adaptive automation. The more applied parts in my work shall first and foremost be seen as tools for building knowledge and testing principles and theories — it is the produced knowledge and insights that are most important.

The initial scope of the F AUTO project represented a high-level idea about the concept of flexible automation in control room environments and how it could be used to improve them from a human-automation collaboration perspective. These ideas represented a scope far too wide for the dissertation, nevertheless, they served as a good starting point. Gradually I narrowed the scope down and started to focus on the actual interaction points between human and automation, which is where human and automation meet and in-

teract. Since I performed this work as an industrial PhD student for LFV, my main application domain was Air Traffic Control, even though some of the studies included elements from Train Traffic Management and Vessel Traffic Service, the other two domains included in the F AUTO project. The following main research question was formulated: ***How can flexible human-automation collaboration be designed to avoid classical automation problems?*** (**RQ**).

To answer the main research question, it was found that the reason for why the automation problems occur must be identified. Furthermore, to answer how to design for flexible human-automation collaboration, a design approach is needed as well as visualization concepts. Thus, three different sub-questions were formulated. First, ***what causes already known issues and problems in human-automation collaboration continue to appear?*** (**SRQ1**), second, ***how can a design approach for flexible automation be formulated that mitigates known automation issues within human-automation interaction?*** (**SRQ2**), and third, ***how can interactive visualization concepts support flexible automation?*** (**SRQ3**).

1.4 Publications

The included papers cover all three sub research questions, with the focus of each paper summarized in the following paragraphs. A graphical overview is provided in Fig. 1.2. Thereafter follows a short presentation of the additional papers.

Paper I (Section 5.1) [60], proposes a novel attention support visualization concept, the Soft Visual Cues (SVC), thus addressing **SRQ3**. The study included operators from all three domains. Likewise, Paper III (Section 5.3) addresses **SRQ3**. In that study, the SVC concept is evaluated in a series of real-time human-in-the-loop simulations with ATCOs.

Paper II (Section 5.2) [59], addresses **SRQ1** and **SRQ2**. The paper's main contribution was the Reduced Autonomy Workspace (RAW) design approach for design of initiation of communication from an automation to a human.

Paper IV (Section 5.4), provides a visualization concept to support the RAW design approach developed in Paper II (Section 5.2), thus addressing **SRQ3**.

Finally, Paper V (Section 5.5) touches upon both **SRQ1** and **SRQ2**. The paper focuses on the cross-domain aspects of flexible automation, which contributes to an understanding of the similarities and differences between the domains, and what implications that might have when designing human-automation interaction for future, highly automated systems.

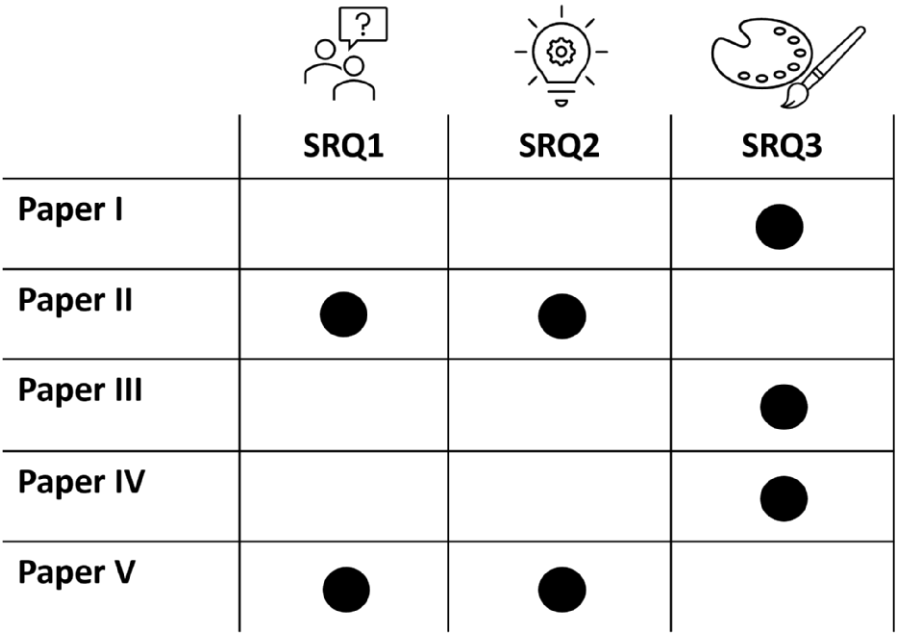


Figure 1.2: The table shows which sub research question(s) that were mainly addressed in each paper.

The additional papers are related to the dissertation focus area, but are either shorter papers or more loosely related to the core topic. The first of these, Paper VI (5.6) [58], relates to the research questions, but it was only a presentation of the overall ideas at a doctoral consortium. The additional Papers VII (5.6) [62] and VIII (5.6) [61] also address F AUTO related topics, the challenge of analysis of entangled control processes and the effect on Air Traffic Controller (ATCO) work strategies by the introduction of trajectory based operations, but they do not directly address the dissertation's research questions. Fig. 1.3 provides a graphical overview of all publications, including additional papers on a timeline. My contribution to the work and how the studies were performed for each paper are described in Chapters 4 and 5.

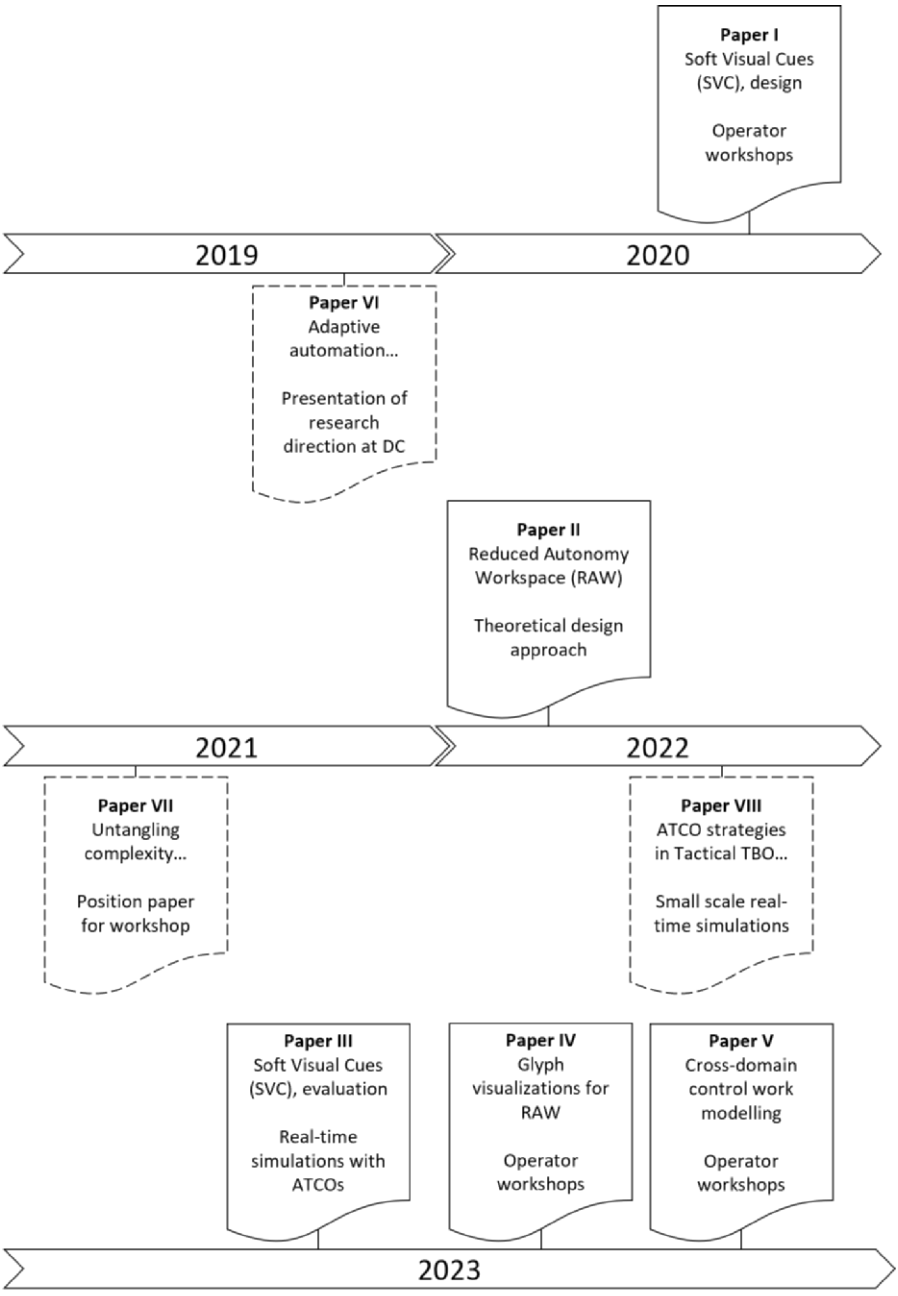


Figure 1.3: A graphical overview of the papers. Solid lines indicate that it is a main paper (Paper I–V) and dashed lines that it is an additional paper (Paper VI–VIII).

1.5 Delimitations

The complete F AUTO concept and the project as a whole include several important and interesting components and perspectives. For example, it comprises the development of sensor technology and the use of it, the use of advanced technology such as AI to interpret the sensor data, and how to make the system understand the overall situation by combining user sensor data with data of the controlled, external processes. Each of these topics could constitute the basis for a dissertation. Although many of these components and perspectives are mentioned, my focus has been the interaction between the human and the automation in the control processes around which they collaborate. Regarding the applied aspects, my work has been focused on the ATC domain, even though the VTS and TTM domains are included to some extent. Likewise, there are many perspectives on automation as such which are interesting and relevant in many aspects, for example, the areas of trust, autonomous behavior of systems, situation awareness, or automation transparency, but those are not the focus of this dissertation.

1.6 Working During Restrictions

A large portion of the work was carried out during the COVID-19 pandemic, which caused challenges and added limitations not usually encountered. As for all people, it meant travel restrictions and for some time also more or less a lockdown, working two full years from home office (from March 2020 to March 2022). It also meant that access to operational premises was strictly regulated. To some extent this affected the time schedule, and it affected the cooperation within the research group which suffered from lack of creative face-to-face meetings, the latter perhaps being one of the biggest challenges. Likewise, the normal social interaction with colleagues was lacking and could not really be compensated with on-line coffee meetings, even though the on-line coffees were far better than not having more informal social interactions at all. Nevertheless, the studies including operators were eventually possible to perform, and for that I am very grateful.

1.7 Dissertation Outline

The main structure of the dissertation consists of seven chapters. Table 1.1 provides a brief overview of the chapters and their content.

Table 1.1: Outline of the dissertation and short presentation of the chapters.

Chapter 1 *Introduction*	This chapter provides a brief presentation of the research project and a broad background to my research. The research questions and an overview of the publications are presented.
Chapter 2 *Project and Domain Background*	The background to the main project and associated concepts is given. The domain of Air Traffic Control is described.
Chapter 3 *Theory Overview*	This chapter provides the theoretical background, mainly focused on automation and related areas, control processes, and domain specific challenges.
Chapter 4 *Methodology and Research Process*	How the work with the different studies was carried out and the main methodological approaches are described in this chapter. It also includes an overview of the technology and prototypes used.
Chapter 5 *Summary of Papers*	The papers, including the additional papers, are presented. Additional work such as posters and other contributions are also mentioned briefly.
Chapter 6 *Discussion*	The results from the overall work are discussed following the F AUTO steps of sensing, interpreting, and adapting. Further methodological perspectives are also discussed.
Chapter 7 *Conclusions*	In the final chapter, the overall work is summarized, and conclusions are presented. As a close-out, there is an outlook towards the future, with suggested areas for further research.

2 Project and Domain Background

The dissertation's perspective on adaptive automation is on human-automation collaboration in control processes, and how adaptive automation can be designed to take advantage of increased automation without falling into classical automation traps or introducing new hazards. New interfaces and principles for human-automation collaboration were developed, and information visualization was used to build knowledge and understanding of the human-automation collaboration. The process of designing the information visualization was an important part of the research, where knowledge was generated both in the form of the resulting designs and in the process of making them.

2.1 Flexible Automation

Throughout my PhD period, I worked within the F AUTO project. F AUTO is an acronym for *flexible automation*, and the purpose of the F AUTO project is to perform research on how flexible automation in control room environments can be used to increase the overall human-automation performance when more automation is introduced. It is a cross-domain project which

includes Air Traffic Control (ATC), Vessel Traffic Service (VTS), and Train Traffic Management (TTM). The cross-domain approach of the project was chosen to allow for experiments and studies in all three domains, either as similar studies in all three domains, or in one or two of them only. Cross domain studies can help to strengthen results — if a concept can be proved successful in more than one domain, it is probably valid in a wider perspective and more likely possible to scale and generalize. If a concept proves to be less successful, then it increases the possibility to gain knowledge about why it did not work. Furthermore, if a concept fails considerably in an early stage in one domain, the others can learn from it. Having that said, my focus throughout the project has been on the ATC domain, which is also reflected in the dissertation as well as in the papers.

The F AUTO Basics

In all three domains included in the project, there are large amounts of data available from sensors looking outwards from the system perspective, such as radar data, flight plan data, and timetables. The trend is an increase of information available in the systems. By combining the external data with information about the human agent's state, focus, and doings gathered by non-intrusive sensors, the goal for F AUTO was to increase the total system performance by adapting the system's behavior and the information presented to the operator. Furthermore, it is crucial that the adaptation is performed in a flexible way. This constitutes the main principles for the F AUTO concept (see Fig. 2.1), and the resulting process can be described as three steps:

Sensing → Interpreting → Adapting

Sensing the operator can be performed in various ways, for example by using eye tracking, or by measuring heart rate variability or skin conductance. In F AUTO, focus has mainly been on eye tracking since it can provide several different types of information. Attention can be derived directly from looking at fixations, while gaze and saccade patterns, blink rate, etc. can be used to assess workload, for instance. By relating gaze to information about the looked at objects, it is also possible to build a more complete picture of the overall situation. Furthermore, there are camera-based eye trackers that are physically non-intrusive, which is probably required if they should be used in an operational environment. Having that said, for the studies we have

used both (1) camera-based, non-intrusive eye trackers, and (2) eye trackers integrated in glasses, which are more intrusive. Experiments and studies in the F AUTO project were mainly supposed to be exploratory with the use of prototyping and user/operator involvement. In parallel, theory has been developed that can be used to model how the adapted automation and human agents collaborate. As concepts were refined, the final goal of the project was larger demonstrations where the concepts should be integrated in an advanced, AI-based, F AUTO environment. Some closing studies are still to be performed, but it is work in progress. In all domains, we started to look at current systems and current automation and then shifted focus towards future systems and automation as the project proceeded.

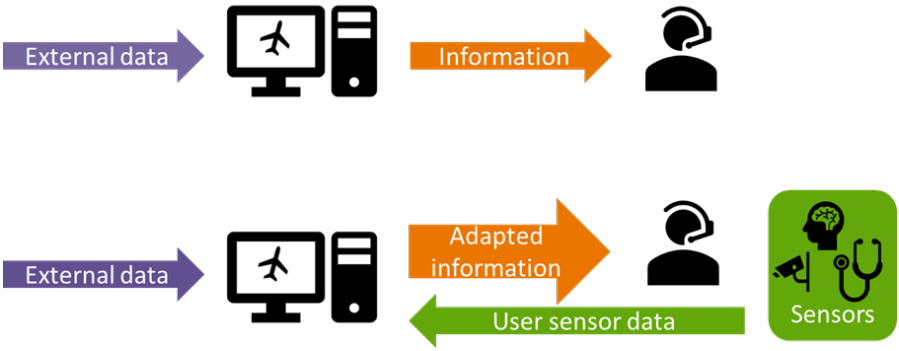


Figure 2.1: High level description of the F AUTO principles. In the upper figure, data is received by the system, processed, and presented to the operator without any regard to the operator's situation. In the lower figure, this is combined with data about the user to create a flexible automation that can provide information and communication adapted to the user's situation, thus enhancing the performance of the joint human-automation team. (Image adapted from the poster teaser presentation at SESAR Innovation Days 2016 (Nylin)).

Control Room Environments

The control room environments from the domains addressed in the F AUTO project have in common that they contain a lot of visual information presented on screens. In Air Traffic Control, it can, for example, include air situation displays with aircraft positions, aircraft labels with information about aircraft's callsign, present altitude, assigned, aircraft type, wake turbulence category, etc. On the same screen, there is information about airspace configuration, prediction of future aircraft trajectories, a list of approaching aircraft, aircraft in holding, tools for measuring distance, tools for conflict detection,

and much, much more. Furthermore, most of this information is dynamic. Then there are other screens for voice communication systems, additional tools, and supplementary information systems needed to perform the Air Traffic Controller's tasks, such as weather information. Fig. 2.2 shows a typical controller working position at an Air Traffic Control Center (ATCC).

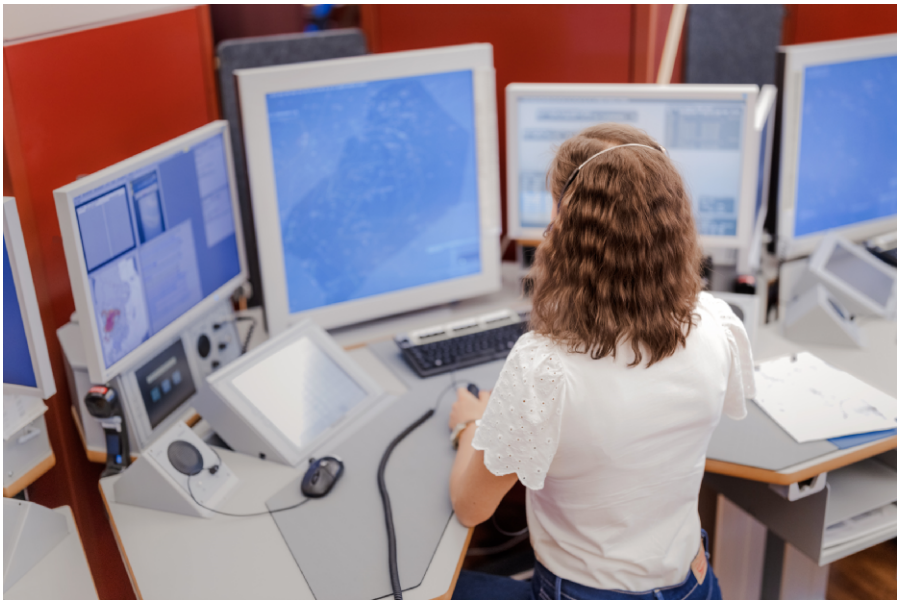


Figure 2.2: A typical controller working position at the Air Traffic Control Center (ATCC). The square screen in the middle is the air situation display with the real-time representation of the air traffic in the controlled sector. The screen on the right contains additional information directly related to the traffic. The screen on the left contains supportive information such as weather forecasts. The tilted screen recessed into the desk is the voice communication system display, and as can be seen, the controller uses a headset for radio and intercom communication. Photo: LFV.

VTS and TTM control rooms share these properties, i.e., they are information-rich environments where information is presented on multiple screens, even though the type of systems and the layout of the control rooms differ. The control room environments have been developed over a long time and the persons using them are highly trained and skilled professionals. Hence, there exists a well-established paradigm for communicating information about the controlled processes and system information to the operators. However, with respect to flexibility, it is still the humans that must be flexible and adapt to the systems and their communication. This knowledge and insight about the control room properties has been important when working on concepts aimed at improvements in current systems by flexible automation, as

well as design of flexible automation for future concepts. Working with visualizations in parallel with the conceptual parts of flexible automation has been needed to proceed with the concepts. On the one hand, if carefully designed, visualizations supporting flexible automation can be used to forward needed information to the operators in smart and efficient ways. On the other hand, badly designed visualizations can lead to high workload and ambiguous communication, not least when complexity rises. If so, it does not matter how good the idea for the flexible automation is on a conceptual level.

Visualizing the Information

The fact that there is already a strong focus on the visual modality in the targeted control room environments could have been an argument to investigate other modalities. There are some studies of how to use sonification as a complement to visual information in ATC, for example, in a basic ATC environment by Ngo, Pierce, and Spence [56], in a radar ATC environment by Rönnberg, Lundberg, and Löwgren [73], and more recently in a tower environment by Elmquist et al. [18] to enhance the awareness of aircraft's positions. While these are innovative and possible good complements to the visual modality, there are also reasons for continuing to focus on the visual modality. First, if a starting point shall be today's control rooms, then it means working with visual representations of information. Second, even if there is room for improvements, there are also many well-established elements and techniques in the current visual environments as they have evolved and been refined over a long period of time, and those should not be disregarded. Third, future systems may provide better possibilities to use the aural modality, but as it is today, much communication is performed via VHF radio and telephone, limiting the room for other aural information (alarm sounds are being used, however). Hence, focus has been on visual information and visualization of real-time data about the controlled processes to support the operators. In my work, the design of visualization concepts has served dual purposes. One has been to design adaptive visualizations that can support the F AUTO ideas and concepts. The other has been to use the design process as a tool to understand the concepts. Working with and trying to understand many of the components in the F AUTO concepts, for example, the interplay between control processes and how operators work, can easily become abstract. The supporting visualizations are more tangible

and can be a good tool for own thought, but also for discussion about the concepts with fellow researchers or study participants.

The Analogy With Humans

When working with human-automation collaboration, it is tempting to use analogies of human-human collaboration. It may be a good analogy to understand the human demands on collaboration and communication, and as a means for discussing an automation's functionality and interaction with humans. However, as pointed out by Shneiderman [83], it is important to keep in mind that regardless of how advanced may the technology be, it is not, and will not be, a human even if it acts seemingly human-like. The rapid development of AI may create even more human-like behavior (if we want to). However, even though AI development has taken some large leaps recently, it seems that there is still some time before we get to a point where we can equate interaction and collaboration between automated, intelligent systems and humans to that between humans. It remains to be seen if we will ever get there, and if it is desirable at all. The analogies may still have a value, though, as a tool for discussion and thoughts, and as long as we are aware that there are differences, it may not be a problem. Rather, it is an opportunity to enhance the team by taking advantage of the strengths of the respective parties. Last, but not least, it can remind us about what extraordinary abilities humans possess that we may sometimes take for granted.

Autonomy

Another concept that may lead to different associations and understandings (as well as misunderstandings) in different contexts is that of autonomy. It is a word that can be defined in different ways in different contexts, and like for automation, there are several definitions, shown, for example, by Müller et al. [52], who made a review on different notions of autonomy in industrial systems. It is also a word used in everyday language, which means that there may be many different understandings of it. For example, autonomy can be seen as an exclusively human quality that should not be attributed to machines [87]. In my dissertation, I use the word in a relatively broad sense, and it is intended to describe to which extent an agent, be it automation or human, can make decisions and act, based on its own competency, knowledge, and jurisdiction without having to consult someone else. When

reaching the limit for what can be done within these autonomy limitations, other agents must be involved. Consequently, less autonomy means more cooperation and communication. In more specific cases, autonomy is used in the meaning of Level of Autonomy in Cognitive Control (LACC) as defined by Lundberg and Johansson [47] — in those cases, it should be evident from the context.

2.2 The ATC Domain

As the domain in focus for the dissertation is Air Traffic Control (ATC), a basic introduction to the domain is due as it cannot be assumed to be common knowledge. Even though it is defined by global standards and rules, ultimately by the International Civil Aviation Organization (ICAO), rules and concepts as well as terminology may differ slightly between nations, not least on the different sides of the Atlantic. Unless stated otherwise, the descriptions below refer to Swedish conditions.

When air traffic started to increase in the first half of the 20th century, it became apparent that the air traffic had to be structured and controlled, which led to the development of ATC. Together with air traffic advisory services, flight information services, and alarm services, ATC constitutes what is called Air Traffic Services (ATS), which in turn is a sub-part of Air Navigation Services (ANS) and Air Traffic Management (ATM) [16]. ATC is a safety-critical, high-stakes domain. That means that failures in safety may have severe consequences, hence, safety is always of top priority. As a domain, it is no exception to the general trend of increased automation. This is driven by a constantly increasing traffic load and complexity, demands from airspace users (mainly airline companies, but also other users such as rescue services, military aviation, and general aviation) to decrease costs and increase efficiency, and at the same time, a demand for maintained or even increased safety. In the recent years, environmental and climate issues have become an important driver for development. Many of these drivers are interconnected — a more efficient ATC can lead to more fuel-efficient flights, which in turn is in line with environmental goals. Automation of ATC is also tightly coupled to digitalization, hence there are issues and challenges not only relating to automation, but also to digitalization. However, while including a lot of advanced technology, ATC is generally still performed in ATM systems with relatively low levels of automation [49].

Different ATC Services

When the work performed by Air Traffic Controllers (ATCO) is discussed, it most often refers to the specific work of separating aircraft while still maintaining efficiency, i.e., the core task of ATC. ATC can in turn be divided into three different branches: area control (ACC, sometimes referred to as en route control), approach control (APP), and tower control (TWR). Area control ATCOs handle all traffic in the upper airspace, which in Sweden includes all airspace above 9,500 feet, as demonstrated in Fig. 2.3.

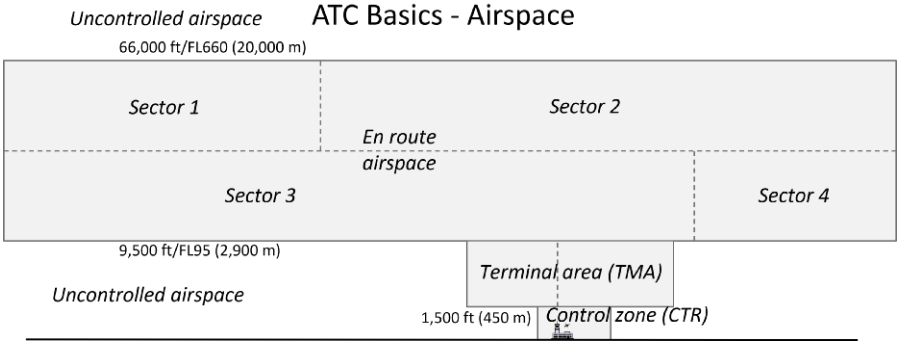


Figure 2.3: A schematic view of the Swedish airspace structure. Somewhat simplified, there are two types of airspace — controlled and uncontrolled. Gray areas indicate controlled airspace in which the separation of air traffic is the responsibility of Air Traffic Controllers (ATCO). The en route airspace covers the whole Swedish airspace. Terminal areas and control zones are controlled airspace around and above airports. In uncontrolled airspace, it is up to the airspace users to ensure separation to other aircraft by obeying the rules of the air and by following the see-and-be-seen principle.

Area Control

The en route airspace is divided into different airspace volumes called sectors. ACC ATCOs often work in pairs, where each pair of controllers is responsible for controlling one sector. The pair consists of one Executive Controller (EC) that communicates with the aircraft and works on a tactical level. The other ATCO, the Planning Controller (PC), also works on a tactical level, but with a slightly longer time horizon; PC searches for possible future conflicts, handles communication with ATCOs in adjacent sectors and other external stakeholders, among other tasks. Depending on the traffic density and complexity, sectors can be joined or split into smaller sub-sectors to maintain a manageable workload for the ATCOs. The air situation display, or radar screen, including support tools presented there, is the main working environment for the ACC controllers.

Approach Control

Approach control is performed in smaller volumes of controlled airspace called Terminal Maneuvering Areas (TMA) between airports and the en route airspace, mainly intended for traffic descending towards and climbing from an airport. As for the ACC ATCOs, the air situation display constitute the main part of an approach ATCOs work environment. For airports with a relatively high traffic load and/or traffic complexity, approach control is often performed by specialized approach controllers co-located with the ACC controllers in an Air Traffic Control Center (ATCC) or at a specific approach control at the airport. For smaller airports, approach control in the TMA may be performed by the tower controller.

Tower Control

Tower ATCOs handle traffic at and in the close vicinity of an airport, the Control Zone (CTR), including ground movements at the airport. Different CTRs may differ in spatial distribution, but a typical CTR in Swedish airspace is approximately 25×10 nautical miles with a vertical distribution from the ground to 1,500 feet. At small airports, there is usually only one ATCO working at a time. At larger airports, multiple ATCOs work together and split up the areas of responsibility, for example, one controller can handle all ground movements at the taxiway system while another handles all movements on the runway and in the CTR. Tower control can be performed from local towers or from remotely operated towers in a Remote Control Central (RTC). The latter is referred to as Remote Tower Services (RTS). However, from a pilot's perspective, it is the same service that is provided. The first airport to be controlled from an RTC went operational in 2015, and currently, RTS is provided to seven Swedish airports from two RTCs.

At the larger towers and ATCCs there are also other roles, such as ATCO assistants or Flight Data Operators (FDO), watch supervisors, etc. There is typically also a large technical organization involved for the operation and management of the different technical systems.

Controlled and Uncontrolled Airspace

The airspace between, under, and over the described airspace volumes is uncontrolled. In total, there are seven different airspace classes, denoted A–G [15], with different requirements on services to be provided and aircraft equipment. In Sweden, only two classes are currently used, class C for controlled airspace and G for uncontrolled airspace. The following description refers to those two classes. In the uncontrolled airspace, separation is the responsibility of the pilots flying there by the see-and-be-seen principle and by abiding the rules of the air. Other air traffic services, such as Flight Information Service (FIS) may still be provided in uncontrolled airspace, though. Information service can also be provided at airports — it is then called Aerodrome Flight Information Service (AFIS). To fly in controlled airspace, there are some requirements on aircraft equipment, such as two-way radio communication and a Secondary Surveillance Radar (SSR) transponder. It is also required to have clearance from ATC. An ATC clearance contains information about how the aircraft is allowed to fly, for example, the lateral route or an altitude. Clearances are also required for taxiing on the ground and for take-off and landings. It is the ATC's responsibility to maintain separation between aircraft in controlled airspace, except for separation between aircraft flying according to visual flight rules (VFR).

Services for Unmanned Air Traffic

A new type of airspace and services for unmanned air traffic (U-space and Unmanned Aircraft System Traffic Management, UTM) are currently emerging. U-space is being designed to meet the demands from the rapidly developing sector of unmanned aircraft, or drones, and is a complement to ATM. However, even though regulations are starting to appear and there is a large number of drones operated, unmanned aviation is still in its infancy. Exactly how it will mature is yet to be seen. What is already clear, though, is that the number of drones and their characteristics make it impossible to treat them as manned aircraft in traditional ATM. The surveillance of drones must therefore follow a different logic than in ATM. Instead of controlling individual drones like an ATCO, the operator may, for example, control traffic flows, services, and constraints.

Provision of Services

The Air Navigation Services of Sweden (LFV) is the main Swedish provider of ANS, and also appointed to develop and provide some U-space services. LFV is a state-owned enterprise and has the appointed responsibility of providing ANS in Swedish en route airspace, at military airports, and at airports operated by the state-owned company Swedavia. At smaller civil airports, the ANS services are open for competition. A unique feature in Sweden is that there is no dedicated civil or military airspace, which enables a flexible and efficient use of the airspace.

The work in F AUTO has mainly focused on the ACC environment and the ACC ATCOs. There are several reasons for that. First, the ACC environment is generally more digitalized and more highly automated than tower environments, thus better suited for studies including interaction with digital, automated systems. Second, area control constitutes a major part of all ATC performed. Almost all commercial flights are flying in en route airspace for a long period, while start and landing constitutes only a small fraction of a flight's duration. In addition, approximately half of all traffic handled in Swedish airspace do not start or land in Sweden, thus only concerning the ACC. Third, the ACC environment has more similarities to the control room environments in Train Traffic Management and Vessel Traffic Service, which make comparisons more relevant.

2.3 ATC, Automation, and Complexity

Increased automation is pointed out as a key aspect for success in future Air Traffic Management (ATM) and Air Traffic Control (ATC) [16] to meet future demands [81]. Nevertheless, the level of automation is still relatively low, and not least, varying between different systems and different areas within the domain [49]. There may be several reasons why this is the case. One reason is the characteristics of the ATM systems as such. They are extremely complex and built in few numbers, thus being very expensive and having long life cycles. Furthermore, they are used in a high-stakes, safety-critical, and highly regulated domain, which puts high demands on proving safety in new systems and likewise for upgrades in the existing ones. Another reason, and the one more relevant for our research, is the nature of the processes to be controlled within these systems. Principles for automation have

largely been developed and researched in domains such as the process industry and manufacturing. More recently, the automotive industry has been a large arena for automation research and development, often striving towards fully autonomous automation. The problem is that principles developed for automation in such domains are not always well suited to be applied directly for automating systems in the ATC domain.

Controlling aircraft in real time is a complex task, but it must be understood what constitutes the complexity, and how this relates to automation. Even though a manufacturing process at an industrial plant may be complicated, it is confined in space; it often includes repetitive tasks; the influencing factors and stakeholder's goals and expectations can to some extent be known; and the operators are in relatively direct control of the processes. This means that predictability is relatively high under normal conditions (failure modes may be a different story, though). Hence, it is suitable for classical automation, which essentially substitutes humans with automation for specific tasks. The process of driving a car is somewhat more complex. The conditions are constantly changing, and it is only possible to predict the future for a limited period as the car proceeds along the road and the conditions change. Automation such as adaptive cruise control or lane keeping assistance in the car must have the possibility to adapt to these changing conditions, and at the same time, it is also affected by the driver's behavior, which may not always be predictable. This requires more of cooperation between automation and human unless the automation acts fully autonomously. However, it is an interplay between one driver and one car, consequently one control process, even though the traffic as a system consists of many of these processes. In ATC, one ATCO (or a pair of ATCOs) control several external processes, the flights, and they do it for a limited amount of time during the flight before handing the flight over to the next ATC unit, about whose intentions and situation they have limited knowledge. The aircraft are in turn flown by pilots, each with their own priorities based on the airlines' policies, such as meeting schedule or minimizing fuel consumption, which are unknown to the ATCO. The aircraft are affected by their environment, such as wind and turbulence, which may or may not be known by pilots and ATCOs. Other parameters unknown to the ATCO, such as mass, may also affect the aircraft performance. As a situation develops, a delay in a decision or the effectuating of it, may change the situation and affect other entangled processes (as discussed in Paper VII, see Section 5.6), both for the ATCO in charge in his or

her area of responsibility and for other ATCOs along the route, as well as for other pilots. This is well illustrated when running real-time human-in-the-loop simulations. The starting criteria are exactly the same for each run, but as soon as the first intervention is performed by ATCOs and pilots, the scenarios develop differently, even if not adding complexity increasing factors such as wind. While the domain is highly regulated, the rules and regulations stipulate the frames to work within, not *how* to achieve the intended results. Added to all this are other factors such as changing wind conditions, experience of the flight crew, which may affect how approach procedures are executed, and much more. The sum of all these factors is what constitutes the complexity in ATC. Table 2.1 provides an overview of some of these factors.

Table 2.1: The table lists some factors that contribute to complexity in Air Traffic Control. The list is not exhaustive, but it gives an idea of why ATC can become very complex. As can be seen, there are a lot of uncertainties and small, but continuous, changes that make exact predictions of future events difficult.

Multiple, simultaneous control processes entangled with, and affecting, each other.
Changing weather conditions in large airspace volumes that are hard to predict.
Unknown preferences and priorities from airspace users, for example, trade-offs between punctuality or minimizing fuel consumption.
Multiple ways of solving similar traffic situations, while the rules and regulations only set the frames.
Continuous small changes are introduced in the processes by things such as variations in response time or when a solution is implemented.
Aircraft performance is hard to predict, hence creating uncertainties in aircraft trajectory predictions.
Each ATCO often controls only one part of the flight and cannot optimize solutions globally.

The aforementioned description of ATC complexity is important for the understanding of why it is complex — it constitutes far more than just solving traffic conflicts between specific aircraft where all factors are known. This may be complicated and requires a lot of skills from the ATCOs, especially if there are many aircraft involved, but it is still a four-dimensional real-time puzzle which can be defined and solved mathematically. The described com-

plexity in the ATC environment is what makes it difficult to increase automation by just letting the automation take over tasks performed by the humans: in which process(es) should an automation work, how should it adapt to the constantly changing situations, and how should it cooperate with the humans in the system with all their expectations, intentions, and goals? And even if it is possible to identify when extended support is needed, what happens when the demand for support is reduced again? The descriptions of control processes and automation in the different domains provided above are short and simplified. However, they show the peculiarities of process control problems in a domain such as Air Traffic Control.

3 Theory

The following sections provide a theoretical background to notions and concepts that have been central to my work. It starts with the presentation of different aspects of automation, with focus on adaptive automation. Known problems and issues with automation are also presented. Furthermore, it includes notes on visualization for adaptive automation followed by analysis and modelling of control processes. Finally, a short background to eye-tracking for sensing operators is provided.

3.1 Automation

Automation in a broad sense has a long history. Hitomi drew up a history of automation in 1994 [31] where he started almost at the rise of mankind, with the first man-made tools. During the industrial revolution, machines, primarily within the manufacturing industry, started to be developed into being operated automatically. According to Hitomi, the term *automation* as we use it today was first used in the mid-1930s in the automotive industry. In a more recent overview, Janssen at al. [36] research the history of, and trends in, human-automation interaction. Like Hitomi, they observe that au-

tomation in a broad sense has been around for a long time, but also how the introduction of computers created new ways of interacting with automation, thus creating a whole new research field of human-computer- and human-automation interaction.

There are different ways of formally defining automation. The dictionary definitions focus on the replacement of human labor, such as Oxford Learners Dictionary (online): "the use of machines and computers to do work that was previously done by people" [65], or Cambridge Dictionary (online) [10]: "the use of machines and computers that can operate without needing human control", while, for example, the International Society of Automation (ISA) has a more technology- and production focused definition: "the creation and application of technology to monitor and control the production and delivery of products and services." [35]. Even though the definitions differ, much of the core meaning is the same. They refer to a technical system, part of system, or tool, made to automatically perform a specific task. As can be seen, some of the definitions, implicitly or explicitly, include *replacing* human labor. It may be the case, but it can also be automation of new tasks that have never been, or cannot be, performed by a human.

Levels of Automation

A common way to describe to what extent something is automated is to use the notion of Levels of Automation (LOA). There are different LOA taxonomies. An extensive overview of different taxonomies was presented by Vagia, Transeth, and Fjerdingen [94]. The notion of LOA as used today was introduced by Sheridan and Verplank [82] in 1978. Since then, the concept has been extended, revised, and adapted, for example, to include different descriptions of levels or a different number of levels (ranging at least from three to twelve levels) [94]. It has also been adopted in industry to describe how automated something is, for example, the six levels of automation defined by Society of Automotive Engineers (SAE) for the automotive industry (see [93] for an overview). A slightly more uncommon approach taken was that of Barbieri, who related the levels to the different industrial revolutions [3]. What the different LOA taxonomies and variants have in common, though, is the basic idea that tasks and responsibilities are split between the automation and the human(s) — to illustrate that, a generic description of LOA is presented in Fig. 3.1. This is also reflected in the literature on adap-

tive automation. Adaptivity is achieved by letting the automation take over tasks to increase LOA to offload the human. It can differ in design, for example, what triggers the automation adaptation [29]. However, LOA does not define what goes on between different levels of automation or between the automation's and human's processes within the same LOA.

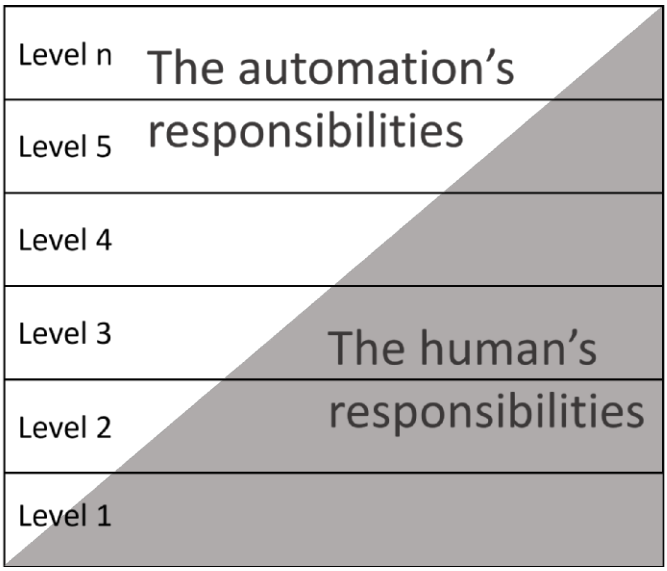


Figure 3.1: There are many different notions of levels of automation. They may have different numbers of levels, and the descriptions of levels may differ. However, they all have in common that they define how tasks and responsibilities, for example decision making, are split between automation and human. The higher the level of automation, the more is managed by the automation.

Adaptive Automation

Adaptive automation in the F AUTO context refers to automation that adapts to a human agent with whom the automation cooperates. The keyword here is *human*. This is an important distinction, as adaptive automation in a broader sense may also include things such a drill that adapts to the current material in which the drill is being operated [13] or responsive manufacturing systems [7]. Parasuraman and Toufik [67] claim that the concept of adaptive automation in the meaning used here, i.e., in human-automation collaboration or interaction, was first introduced by Rouse [74] in 1975. The precise date for the introduction of adaptive automation can surely be debated, but it seems to be safe to say that it has been around for 45 years plus at this point. It is important to distinguish the concept of *adaptive* automation

from the related concept of *adaptable* automation where the automation is actively adapted *by* the human agent, but is not adapting itself (see, e.g., [12, 45, 53, 94]). Adaptive and adaptable automation are not mutually excluding concepts, as one and the same system can contain both, but my work has been focused on adaptive automation.

When performing an overview of the literature on adaptive automation, it was noted that there was steep a increase in publications on the topic around the year 2000. This was reflected upon both by Freeman et al. and Prinzel et al. [27, 69] who noted that around 2000, not very much had been done in the field of adaptive automation, although Prinzel et al. claimed that it was discussed a lot.

Adaptive Automation in ATC

During my literature review on adaptive automation, I noted that a number of the publications mentioned Air Traffic Control (ATC) as a possible domain for adaptive automation, or referred to other studies within ATC, not least this was the case in the then most cited publications related to adaptive automation [9, 37, 51, 68, 97]. This indicates that ATC is a domain where adaptive automation could be important or useful. It does not, however, mean that there is a large quantity of literature on the subject. At the time when the initial review was performed (early 2019), 14 publications relating directly to adaptive automation in ATC were found, and just a few publications have been added to the list since then. If that is few or many articles can be discussed. On the one hand, ATC can be considered a small domain given the number of people directly involved; on the other hand, it is a global domain and has a deep connection to the aviation domain in general, while being a safety critical domain. My conclusion is that there is still a lot to explore regarding adaptive automation in ATC.

3.2 Issues With Automation

According to the F AUTO principles (see Section 2.1), it is the automation that shall adapt to the human operator, and it shall do so in a flexible manner. In order to achieve this flexible automation that we aim for, the understanding of the human operator is a must. Poorly designed automation, adaptive or

not, can cause unintended and unwanted side-effects. It can, for example, lead to the loss of situation awareness (SA) [19, 46] and out-of-the-loop problems [20, 23], automation surprises or mode errors [77, 80], or problems like Bainbridge's automation ironies [2], as summarized in Fig. 3.2. Klein et al. identified several challenges with making automation a part of the team that should be addressed, for example, regarding predictability and communication of status between the automation and the human [41].

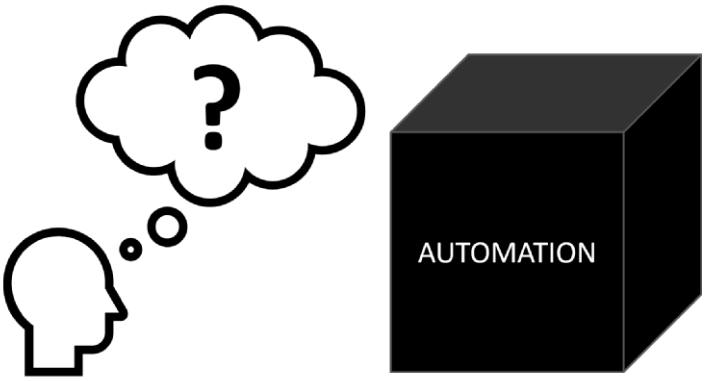


Figure 3.2: Does the operator know what the automation is doing, how it works, or is intended to work? Or is the automation a magic black box that works until it does not and then leaves the problem to be solved by the human? And does the operator need to know how the automation works for to use it? A successful collaboration between human and automation requires a good design of the automation to keep the human in the loop, to make the human aware of the mode of the system, and to avoid that the human is forced to take over when the automation can no longer handle the situation.

One of the probably most well-known and cited descriptions of issues with automation and design of automation is the ironies of automation defined by Bainbridge in 1983 [2]. It describes how potential gains from automation can, ironically, give the opposite effects. For example: designers try to eliminate human errors by automation, instead (human) design errors in the complex automation are introduced, which requires the operator to deal with things that were not thought of or not possible to automate. Another irony is that the more automated a system is, the more training is needed for the operators to maintain their manual skills, which may still be needed unless the automation covers all possible situations. Or, if there is a lack of trust in the automation, if the automation is poorly designed, operators may end up both making decisions about the controlled processes and monitoring the automation, leading to more rather than less work. Although formulated forty years ago, they have been found to still be relevant [4, 21, 88].

Another issue that may appear in human-automation collaboration is when the automation behaves in a way that was not expected by the operator. Sarter, Woods, and Billings [80] described how these automation surprises can occur when more and more automation is introduced. This work in turn related to the previous work by them and others on the issues related to aviation and human-automation collaboration in the cockpit, such as Wiener's paper about glass cockpits [96], or Sarter's and Woods' papers on pilot-cockpit automation interaction [77, 78, 79].

Experiencing automation issues such as automation surprises may result in the loss of situation awareness (SA) and being thrown out of the control loop, not least when the level of automation increases [20]. Endsley [22] emphasized that to maintain SA, the operator must both know about, be able to interpret and understand the current situation, and to extend that knowledge in time to be prepared for what is coming. Much of the research on SA has its origins in the aviation domain, and there are several related theories and models, for example by Salmon and Boudreaux [75], Lau, Jamieson and Skraaning [43], and Lundberg [46]. A similar problem is what may arise when the level of automation is reduced and the human operator must regain (more) manual control, a problem addressed, for example, by Kaber and Kim [38], who studied the use of auditory cues for control mode changes.

3.3 Visualizations for Flexible Automation

Enhanced human automation-collaboration by flexible automation requires interfaces for communication between automation and human operators. In this work focus has been on visual communication by information visualization. Spence [86] and Card [11] provide reference models of information visualization that describe how data is represented by a visualization in a visual structure, which is in turn presented to (e.g., on a display), and interacted with, by a user. These are generic models that include interaction by the operator as feedback to the system. In the F AUTO context, the input from the operator's interaction is complemented by data about the operator, for example, regarding attention.

My work has included both modification of an existing visualization design by a new concept, Soft Visual Cues in Papers I (Section 5.1) and III (Section 5.3), as well as the creation of a new one, the glyph-based design for communication initiation in future F AUTO systems in Paper IV (Section 5.4). Glyphs come in a wide range of formats and variants. Those considered relevant in this context (for presentation in an ATC graphical user interface) were 2D glyphs. What they have in common is that they are symbols that visualize data through their properties, such as shape, color, or position. Examples of common glyph types are bar charts, rings, whisker plots, pie charts, star glyphs, or petal glyphs [28, 40, 95].

Specific Visualization Challenges in ATC

A choice was made to concentrate on the visual modality (see Section 2.1). Even though the visual modality is widely used already in the ATC control room environment, and one must be careful not to overload it, it also means that there is already an elaborated visual, on-screen information presentation that the ATCOs are trained for and used to deal with. ATM systems contain large amounts of information, both about the main processes of controlling aircraft and different supporting processes. This large body of information must be processed and communicated in real time. The ATCOs' ability to perceive information is limited, both with respect to available time, which requires good timing, and amount. This differs from, for instance, most of the examples brought up by Spence [86]. The same trend was identified when reviewing literature for Paper IV (Section 5.4) about glyphs — most visualizations were related to post-analysis of data. The available screen space to present information on is also limited. The importance of time is not limited to timing of the communication, it is also important with respect to the communication of temporal aspects, such as how much time there is left until two aircraft are coming too close to each other. Hence, information visualization of communication between automation and ATCOs must be space efficient, easy to perceive and interpret while still containing all information needed, and in some cases also be capable of communicating temporal information.

Attention Support

As already mentioned, there is already a lot of information presented in the ATCO working environment. For the Soft Visual Cues (SVC) concept for at-

tention support, an existing visualization of conflicts between aircraft was used, presented in Paper I (Section 5.1). It addresses the problem that information can easily be missed [48, 49], even if it is important. As pointed out by Roda [72], there are many aspects of attention support and why information can be missed, for example, that the context must be considered. It must also be based on some knowledge about human perception, for instance, the risk of change blindness, the fact that seemingly prominent changes in a familiar environment can nevertheless easily be missed [84, 85]. ATCOs are trained to scan areas of interest and if it can be detected that a certain object has not been looked at, it can be assumed that it has been missed given that the performance of the eye tracker is sufficient. It might have been intentionally neglected, though. The SVC is supposed to give attention support where this is not known. Differences in some properties of graphical objects, for example, the rotation of a square or one dot in different color from other dots, are very quick for humans to detect. This phenomenon is called preattentive processing [98] and was is an important foundation for the SVC concept together with the known gaze scanning patterns of the ATCOs.

3.4 Describing and Analyzing Control Processes

There are several approaches for how to describe and analyze work- and control processes. For example, a Cognitive Work Analysis (CWA) can be used to identify relations between events at different levels of abstraction, roles and responsibilities, constraints, etc. [54, 55, 70], or to compare different complex sociotechnical systems [50]. However, a CWA is focused on relations, not processes, and lacks descriptions of temporal aspects. Furthermore, while the CWA maps of relations may be exhaustive, it also means that they can be extremely complex, and there is a risk that it ends up with a description where there are relations between virtually all mapped entities, thus giving little guidance for how a system works, or should work if used to design a new system. The Contextual Control Model (COCOM) is a narrowed down model that focuses on the functions specific for one control process and how it is performed in different control modes. It ranges from the strategic mode where several goals can be addressed and there is time to make informed decisions, to the scrambled mode, where the time for considering alternatives

is inadequate and actions randomly implemented [33]. The COCOM was extended in the Extended Control Model (ECOM) with four layers of activity [33]. Though Hollnagel and Woods emphasize the importance of time and how, for example, the ECOM includes temporal aspects, the models have circular representations of reoccurring patterns. A similar approach is taken in Pacaux-Lemoine and Flemisch's model for cooperative control [66] — it follows a pattern where information is gathered, analyzed, and acted upon following a decision, and then it starts all over again. Another alike pattern can be found in Boyd's OODA loop (Observe, Orient, Decide, Act) [5, 71] and the Dynamic OODA loop (DOODA) [6]. Hence, these models provide a formalized way to model a general principle, but not what is going on in the control processes over time or how the processes relate to and affect each other. Neither do they give any directions for a designer with respect to how a system should work with temporal aspects. Fig. 3.3 illustrates the issues of modeling temporal aspects in circular models.

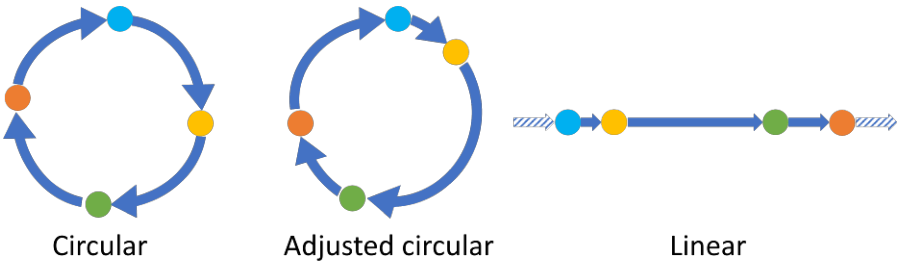


Figure 3.3: When modelling control process events, a circular model can explain the order of the events while it illustrates a principle. If the actual time between the events shall be modelled, an adjusted circular model would be needed. However, it still only illustrates one cycle of events. To be able to illustrate, and to understand, the distribution in time between certain control process events and to allow for multiple events over time to be modelled, a linear representation is more well-suited. This also allows for different patterns of events to be put together into longer episodes with different patterns of events.

In ATC, temporal aspects are crucial to understand the effects of different actions and why a situation develops in a certain way. Likewise, temporal aspects are important to be able to model and understand how an automation in the ATM system works and the interaction between the ATCO control processes and the automation's control processes. One framework that incorporates the processes' extension in time and enables modelling of temporal aspects is the Joint Control Framework (JCF), which builds on previous works, like the aforementioned ECOM and COCOM models, and other similar models and frameworks [47].

3.5 The Joint Control Framework

The use of the Joint Control Framework (JCF) [47] has been an important part of my work, not least for the development of the Reduced Autonomy Workspace concept where the possibility to model temporal aspects was a key factor. The JCF combines the notion of different levels of interaction between subject and objects in control processes from different previous frameworks. It builds upon and brings together previous frameworks for process control and uses six Levels of Autonomy in Cogntive Control (LACC), where the highest level, *Framing*, contextualizes the process interactions. One of the most prominent features of the JCF is that it provides the possibility to analyze interaction over time, how different processes affect each other, and how information flows between processes. The framework is visualized as a score, quite like a sheet music score, although with six lines instead of five. The temporal dimension is simply the horizontal extent of the score. Each line represents a LACC at which the interaction between a subject and an object takes place, for example, an ATCO interacting with an aircraft. Each interaction point is denoted a joint. As a minimum there are three types of joints: Perception Points, Decision Points, and Action Points, see Fig. 3.4. The notion can be extended if other types of joints are needed to describe the interactions. Several scores can be used to visualize multiple, parallel processes.

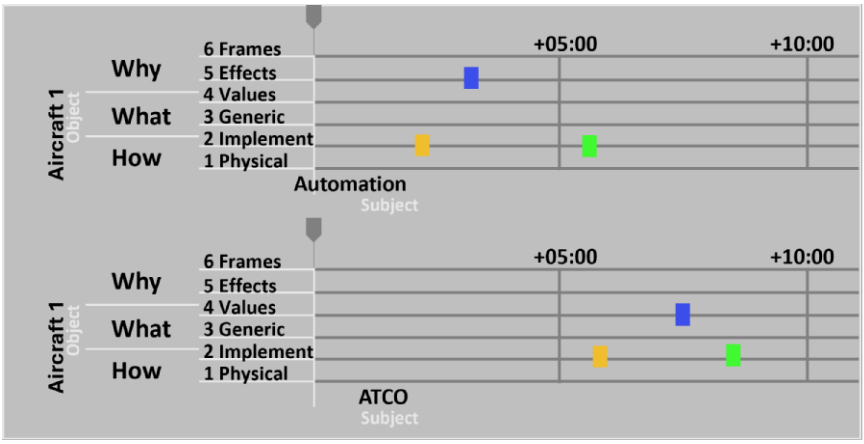


Figure 3.4: A very basic example of two control processes visualized using JCF scores. It depicts an automation (upper score) and an ATCO (lower score) that both interact with an aircraft. The lines show the level of autonomy in cognitive control. The different colors of the joints depict the type, from left to right: orange = perception, blue = decision, and green = action. For a more conceptual description, see Lundberg and Johansson [47, p. 387].

In complement to being a powerful framework for analysis and comparison of control processes, it can also be used as a tool for designing new concepts, as was done for the RAW design approach [59]. During my PhD period, a tool, the JCF Editor, was developed by a team of engineers at Linköping University. The tool made it very easy to quickly draw JCF scores and increased the usability of the framework significantly. It includes a video player where recordings to be analyzed are synchronized with the score. It also includes the functionality of marking specific episodes, noting relations between joints, and adding text descriptions to the joints.

3.6 Sensing the Operator With Eye Tracking

The focus on the visual modality in the targeted environment in combination with the need for operator tracking in the F AUTO concept made eye tracking the first choice for gathering data about the operators. In my studies, it was used for attention detection, see Fig. 3.5. Regardless of which measures one uses from the eye tracker, one advantage with eye tracking is that it is possible to relate it to what is looked at, i.e., what is displayed at the screen. This information can in turn be used to create a direct connection to the controlled processes, which is an important F AUTO aspect.

Eye tracking has been used in research since the mid-1900s. However, early eye tracking technology was far from easy to use and often demanded participants to sit very still. Neither was the technology very common, and up to the 1980s, researchers often had to build their own devices [34]. An early example of an eye tracking study (not ATC-related, though) is the work of Hillsman et al. in 1970, who demonstrated how eye tracking can be used, including the use for input to a simple video game [30]. Looking at published papers within the ATC domain, eye tracking for use in ATC control room environments have been used since around year 2000, even though there are a few earlier examples, such as Thackray and Touchstone from 1988 who used video recordings to assess eye and head movements [91]. More recent examples of studies within the ATC domain where eye tracking has been used show a variety of different questions addressed. Some examples of this are a study on the impact of conflict angles [17], identification of visual scan patterns for conflict mitigation [25], assessment of ATCO workload to-



Figure 3.5: An example of how modern eye-tracking equipment can look like today (from the Soft Visual Cues Study in Paper III). In this case, it is a technology based on infrared sensors built into the frames of the glasses, thus enabling tracking regardless of where the user looks. The drawback of this solution is that the gaze cannot directly be related to on-screen objects without using additional tools, and it also hinders participants from wearing correctional glasses.

gether with electroencephalogram (EEG) [89], and a study of alerting designs to reduce ATCO response time to alerts [39]. Other applications include eye tracking for input to ATCO support tools [63] and for attention guidance [64].

4 Methodology and Research Process

The overall theme in my work has been flexible automation. My focus has been the automation's interaction with the human operator in the system and the human's experience of this interaction, rather than focusing on the automation itself. The nature of the overall research process has been exploratory and pragmatic, and it started with a relatively wide and open scope. The studies performed have included both qualitative (Papers I, IV, and V — see Sections 5.1, 5.4, and 5.5, respectively) and mixed method approaches (Paper III and to some extent Paper VIII, see Sections 5.3 and 5.6). The studies have mainly consisted of operator workshops and real-time human-in-the loop simulations. Two of the papers, Paper II (Section 5.2) and VII (Section 5.6), were purely theoretical studies. Paper VI (Section 5.6) was also mainly theoretical, even if it included some early prototyping, see Fig. 4.1 for an overview. Framing the research from a design perspective using Fallman's interaction design research triangle [24], I consider my work to relate mostly to the design studies area in the sense that my focus has been to build knowledge and to, together with operators in simulations and workshops, explore how the concepts and visualizations perform. This focus on using the design process and studies of the resulting artifacts and concepts probably could also define the studies as research through design [26,

99], or as constructive design research [42], where the latter emphasizes the construction of something as part of the building of knowledge where the artefact itself is considered important.

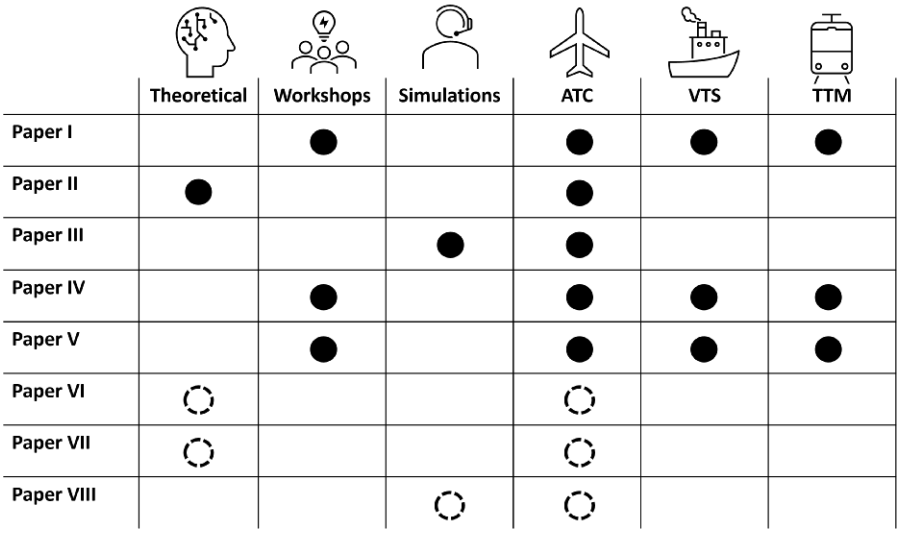


Figure 4.1: The figure illustrates which papers, including the additional papers VI to VIII, that were mainly theoretical or included empirical studies in the form of real-time human-in-the-loop simulations or workshops. It also illustrates which domains they addressed.

The sections below describe the process from scoping and narrowing down the research area. It furthermore describes the iterative development of the different concepts and visualizations, how operator involvement was achieved, and finally about technology and prototypes used in the studies. In addition to more formalized methods, a lot of time has been spent on thinking about the ideas and concepts, and on informal discussions to get different perspectives and input on my work.

4.1 Scoping

The F AUTO project had a relatively wide scope. It was aimed at flexible automation in current and future control room environments and included, but was not limited to, Air Traffic Control (ATC), Vessel Traffic Service (VTS), and Train Traffic Management (TTM). Hence, the first step was to define and narrow down a smaller scope for the dissertation work. My assignment in

the F AUTO *project* was to lead the ATC part. However, ATC is by itself a large area. With respect to ATC control room environments, it can, irrespective of services provided, roughly be divided into tower environments and radar environments. A tower environment normally also contains a radar screen for approach services or as extra information for the tower ATCO, but it does not constitute the main part of the environment. As discussed in Section 2.2, several reasons led to the decision to focus on the area control radar environments. First, it shares more control room layout properties with VTS and TTM than the tower environment. Second, it is highly digital, thus making it easier to work with information visualization. Remote and digital towers are indeed digital as well, however, there were already many projects going on related to remote towers, which meant potential less room for novel research contributions at that stage. Furthermore, even if they are digital, they are still very similar to a traditional tower. Third, with an educational background within area control and experience of working with simulation and validation in radar environments, it was possible for me to get into the actual research questions more quickly. In addition to the educational background within ATC, I also have more than 20 years of experience working in the domain, mainly within research and development.

While flexible automation is intended to go beyond the concept of adaptive automation, the latter is nevertheless a substantial part of flexible automation. A literature review on the state of the art in adaptive automation was performed early in the project as a part of scoping the research question.

4.2 Concept Development

Concept development has followed an iterative cycle with internal discussions and informal workshops within the research group. Not least was this the case for the development of the Reduced Autonomy Workspace (RAW) concept, which included many hours of discussions and sketching of ideas, some examples are presented in Figs. 4.2 and 4.3. Generally, the procedure was that I worked out an outline of the ideas in the form of simple sketches which I presented to the other members of the research group. This was used as starting point for the discussion within the research group, and the outcome provided input for the next iteration.

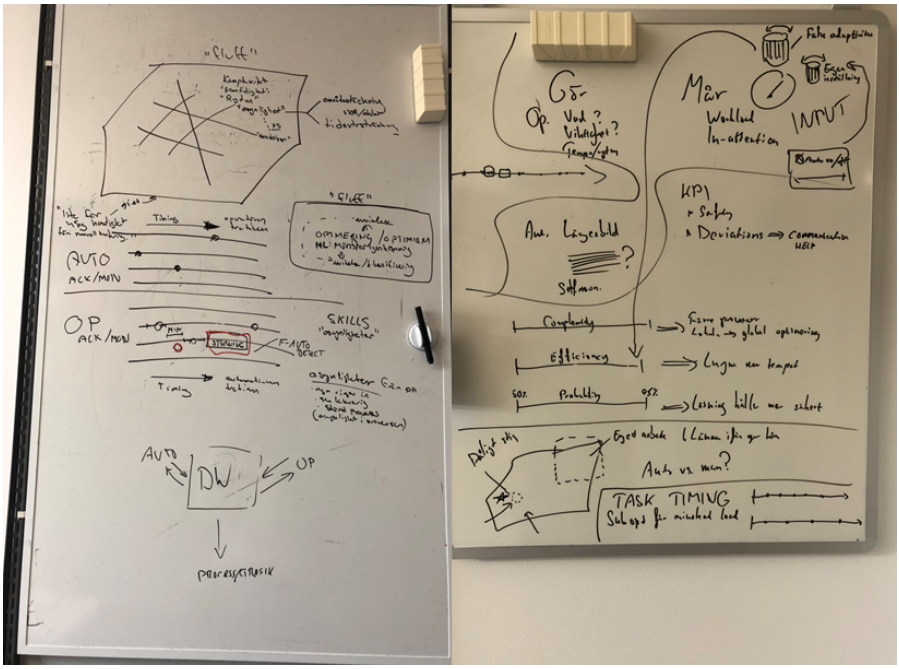


Figure 4.2: White board sketches in an early development state of the Reduced Autonomy Workspace. On the left whiteboard, one can catch a glimpse of two control processes sketched in a simple Joint Control Framework notation (the parallel lines with dots on them depicting different perception, decision, or action points).

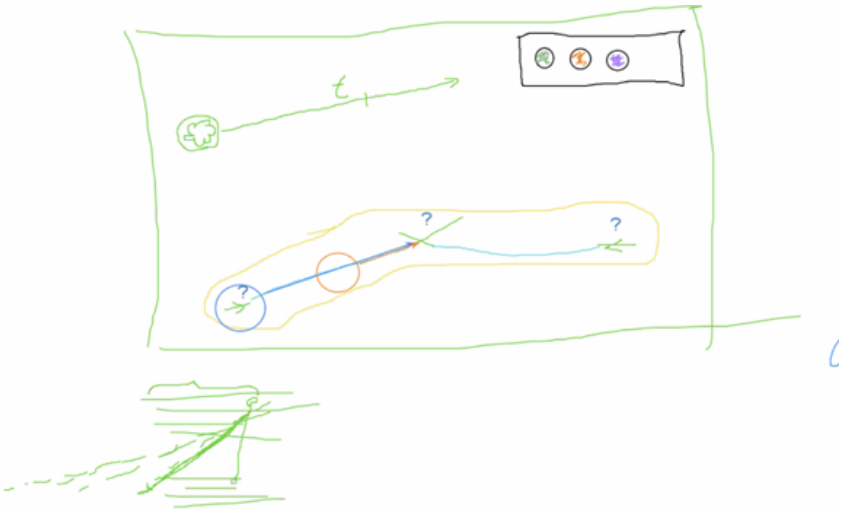


Figure 4.3: Working with visualization ideas for the Reduced Autonomy Workspace. Sketch of a traffic situation with two control processes, an interaction GUI (glyphs) in the upper right corner. The sloped, thicker line between the work processes depicts information exchange over time, the basic idea upon which the Reduced Autonomy Workspace is built.

Parallel to the process of sketching theoretical concepts, I was also sketching visualization concepts and designs. The visualizations have served two main purposes. First, they were needed to put the theories to test, not least in the work with the Soft Visual Cues (SVC) where evaluating the designs was just as important as exploring the visualization concept. Second, the process of designing the visualizations has in itself been important for the development of the concepts. Most of the time it has been an iterative process where a conceptual F AUTO idea emerged, simple sketches were made on paper or basic drawing tools, the sketches gave new insights on the concepts, the concepts were revised, and so it continued.

For Papers I and III (see Sections 5.1 and 5.3, respectively), the idea of using the Conflict and Risk Display (CARD) emerged from studies of incident reports and previous research on the MTCD/CARD [48]. It also included feedback gathered at an external workshop with LFV safety assessors, where the participants brainstormed around areas where attention was known to be an issue.

4.3 Operator Involvement

Operator involvement has been a cornerstone in the F AUTO project, and the work upon which this dissertation is based is no exception. However, there are a limited number of ATCOs, and they are often wanted as participants in various project. Furthermore, they are of, needed in the production of air traffic services. In total there are between 500 and 600 active ATCOs at LFV [44], covering all ATC branches of tower (TWR), approach (APP), and area control (ACC). Therefore, it must always be a deliberate decision to include ATCOs in a study, or, if alternatives such as including ATCO students shall be considered. However, when the expertise and experience of ATCOs are needed, there are not many suitable alternatives. On the positive side, the group is relatively homogeneous when it comes to their professional role — they have been selected to be controllers by the same criteria, they have all gone through the same training, including both basic training and continuous competence/assurance training, and they all perform the same tasks together on a daily basis. To some degree this ensures that each ATCO can be considered representative for the community, which makes even small studies valuable. Having that said, I am fully aware that there are also differences in how individual ATCOs prefer to perform their work in different sit-

uations. As I am fortunate enough to have more than 20 years of experience from the ATM domain myself, including an ATCO education with specialization in ACC, I have been able to use that experience in the development of the concepts and focus the ATCO participation at the evaluation of the concepts. Operators from the TTM and VTS domains have also been involved in some of the concept evaluations. Like the ATCOs, they are also part of small communities with limited possibilities to participate in activities outside their operational duty. In all cases, I have managed the recruitment of participants through the resource allocation functions at the respective organizations with some help from F AUTO research group. The participants have then been scheduled to participate in the studies as a part of their everyday work.

Workshops

Three of the papers were based on data gathered in workshops. For the study in Paper I (Section 5.1), the workshops included ATCOs, VTS operators, and train dispatches. The participants evaluated different designs proposals for SVC presented in a simple prototype. Even though it included participants from all three domains, most focus was on the ATCO workshop. The glyph paper, Paper IV (Section 5.4) included three separate workshops, one per domain. The visualization concept was discussed during the workshops, where the participants contributed to identification of information types to be included in the glyphs. The same evaluation of the reading of temporal aspects in the proposed glyph design was performed in all three domains. Finally, Paper V (Section 5.5) was also based on the results from operator workshops including all three domains. The first workshop included identifying traffic situations common to all the domains. In the second workshop, the participants jointly (per domain) performed an initial Joint Control Framework (JCF) analysis using post it on a whiteboard, see Fig. 4.4. The ATC part of the workshop was based on a similar case used as a typical case for the missed conflict information in the Soft Visual Cues. The workshops are described in more detail in the Summary of Papers (Chapter 5).

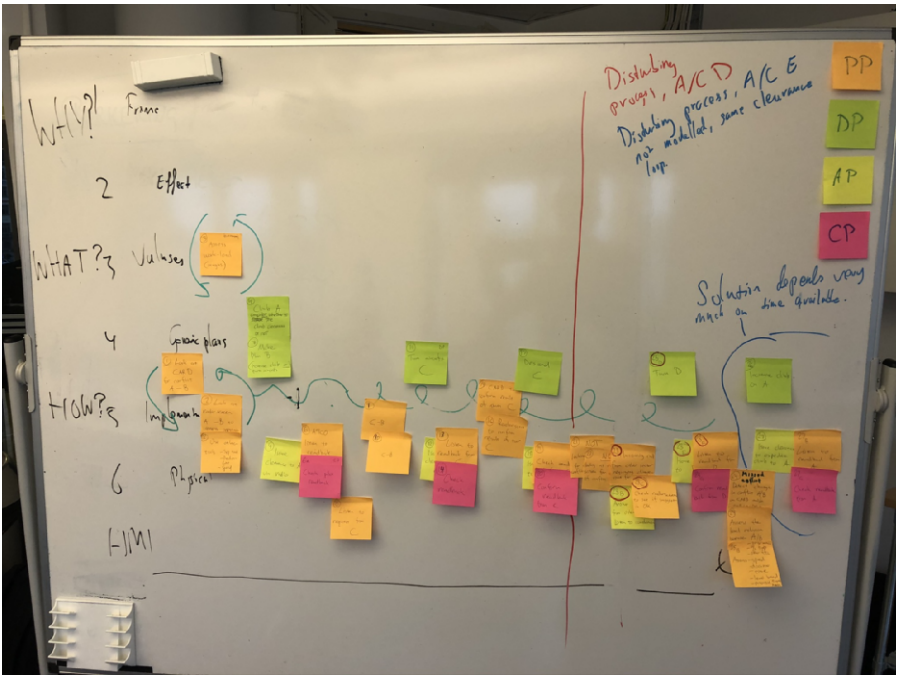


Figure 4.4: Joint Control Framework analysis performed by operators on a traffic situation. Each post-it is a joint in the joint control framework notation. The joints are either perception points (orange), decision points (green), action points (yellow), or confirmation points (pink). The vertical position of each post is a rough indication of level of autonomy in cognitive control for the joint, and the horizontal position is the distribution over time. What is modelled here is the process of an Air Traffic Controller (ATCO) working with a certain conflict between two aircraft, where other processes interfere.

Real-Time Human-in-the-Loop Simulations

In those cases where concepts reached a higher level of maturity, real-time human-in-the-loop simulations were performed. Two such simulations have been carried out. The first one was for the additional paper, Paper VIII (Section 5.6) [62], where a tool for tactical trajectory-based operations (TBO) was developed and tested during a small scale simulation (two days with one ATCO each day). The second one was a larger simulation with respect to the number of runs and participants, five days and nine ATCOs, which resulted in Paper III (Section 5.3). Both simulations were run in a setup consisting of a single ACC sector supported by one simulator/pseudo pilot who operated the simulated aircraft and a rest-of-the-world (ROW) position to simulate adjacent sectors. The ACC sector was a real sector in Swedish airspace and the traffic scenarios used were based on real traffic data, although slightly modified to ensure the appearance of conflicts, which were of interest in both

studies. The validity of the traffic scenarios was confirmed by the participants.

The simulations were performed using LFV's research ATM simulator, Narsim [57]. Originally developed by the Netherlands Aerospace Centre (NLR), it has been used by LFV for 15+ years to perform research within ATM with real-time human-in-the-loop simulations. The simulator is designed as a generic ATM system, but it is highly configurable and has been adapted to mimic the operational ATM system for area control ATC in Sweden, TopSky. It also has the ability to simulate tower ATC. Communication between pseudo pilots and ATCOs is performed through a simulated radio communication system. The aircraft's behavior is determined by physical flight models, which take parameters such as aircraft mass and cost index settings into account. Realistic conditions such as wind can also be simulated.

4.4 Prototypes and Technology

To put the ideas developed in this dissertation to test, some new technology and prototypes have been developed. For each of the studies, the question has been asked: which level of fidelity is needed? ATM systems are inherently complex, and there is always a risk to get stuck in technical development, only to realize that the same results, or even better, could have been achieved with less sophisticated, but fit for purpose technology. Bill Buxton [8] refers to this as finding the *right* fidelity — higher fidelity is not always better. In Paper I (Section 5.1), simple PowerPoint visualizations were used as prototypes. I developed early software prototype sketches in Python, an example of how it looked like is provided in Fig. 4.5, but they turned out to be quite complex to build without adding any value, as the only purpose was to evaluate the design as such. For a skilled software developer it might have been easier, but I was learning Python programming at the same time, so sufficient quality and stability were not foreseen to be achieved within a reasonable time frame. For the first Soft Visual Cues study, it was the comparison between different design candidates that mattered, not a real-time evaluation of the concept. However, the work of writing the code was not wasted as it provided value by sparking creativity in the design process, not least when specifying the prototype for the follow-up study on Soft Visual Cues, Paper III (Section 5.3).

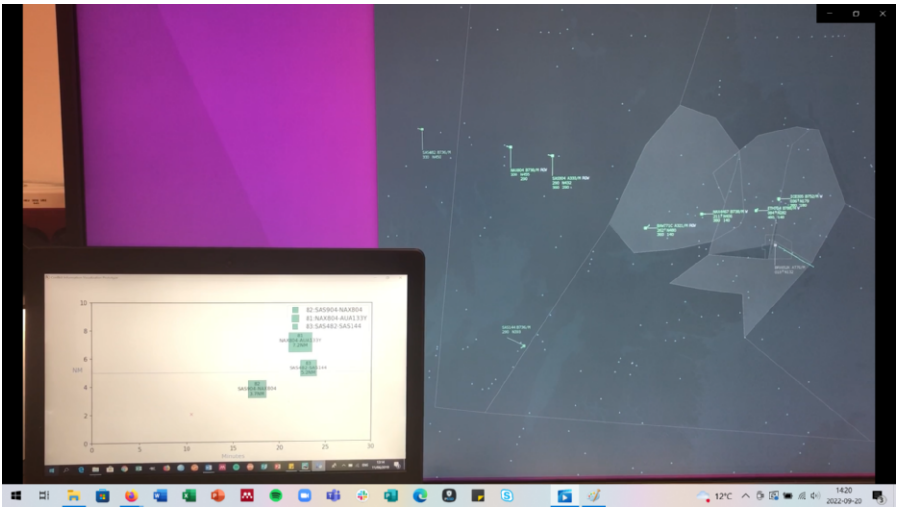


Figure 4.5: The first, draft prototype of the Soft Visual Cues made by myself is shown on the left, white screen. While fed with data from the simulator, it was run on a separate computer, on a separate screen, and was found to be too unstable to use for simulations. A Tobii X3-120 eye-tracking device (not visible on the image) was used for input. It also turned out that some property changes were much easier to implement than others. The image demonstrates how size encodes the property change.

The Soft Visual Cues System

For Paper III (Section 5.3), the concept of Soft Visual Cues had reached a level of maturity where it was ready to be tested and implemented in a realistic setting. This led to the development of an SVC system where eye tracking was combined with real-time analysis of a radar screen to identify objects on the screen that had been missed. The prototype was developed by a team of software engineers at Linköping University. For this study, the focus was on visualization of conflicts between aircraft in an ATM system. A conflict in this respect is when two aircraft are predicted to be close to, or below, the stipulated minimum separation. The conflicts are detected by a type of tool called Medium-Term Conflict Detection (MTCD) that compares aircraft's flight trajectories. (The tool in the simulator used is called Aircraft Conflict Detection system, ACOD, but it is similar to the MTCD used in operational systems.) Each conflict is visualized as squares in a separate window by a conflict visualization tool, called Conflict Window in the simulator and Conflict and Risk Display (CARD) in the real ATM system. The squares representing conflicts are drawn in a scatterplot with closest predicted distance between the conflicting aircraft on the y-axis, and the time to when it will occur on the x-axis,

as presented in Fig. 4.6. To help the operator to quickly assess the urgency, 5 nautical miles distance (minimum separation) is marked in a darker shade.

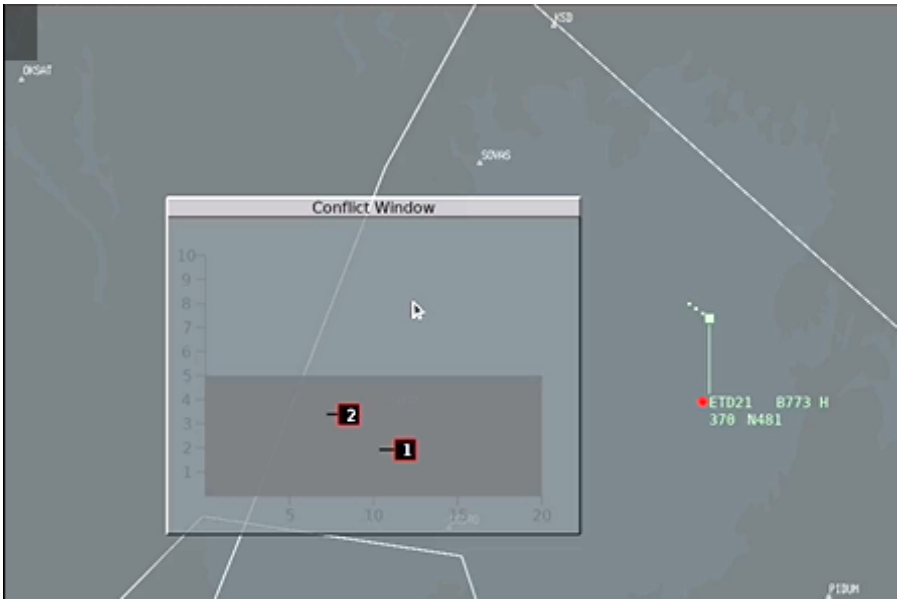


Figure 4.6: The image shows the Narsim Conflict Window. The two red-framed squares represent two different conflicts. Time to shortest distance between the aircraft is on the x-axis and shortest calculated distance is shown on the y-axis. The darker shade of gray in the lower part of the scatterplot depicts 5 nautical miles, which is the minimum required radar separation.

The ATM system used in this case was LFV's ATM research simulator Narsim [57], but similar functionality is used in the live ATM system, TopSky [92]. The layout is conceptually similar, but with a slightly different design, for example, regarding the design of the conflict symbols. The soft visual cues in the form of a change in geometry (rotation) of the conflict visualization boxes was implemented as an overlay in the image before presenting it on the screen, i.e., a real-time modification of the image. The system could not only identify the boxes, but also the ID (serial numbers) presented in each box. By this, the system could keep track of the different boxes if the position of the boxes or the whole window changed position. The ID numbers are also used for the operators as a help to identify the conflicts. The positions of the boxes were updated and moved if the predicted shortest distance, or predicted time for occurrence of shortest distance, changed. The user could choose freely where on the screen to place the window. As long as it is possible for the system to identify a certain object on the screen, the technology could be used for any on-screen object. This technical solution meant

that no integration with the simulated ATM system was needed as analyses and modifications were performed directly in the image, which reduced the complexity of the overall setup.

A Tool for Working With JCF Scores

The analysis of work processes using the Joint Control Framework [47] was an important part of the work that resulted in the Reduced Autonomy Workspace concept, first presented in Paper II (Section 5.2). However, visualizing the processes in the score notation using simple drawing tools in Office programs was not feasible. The same conclusion was drawn in most projects working with the framework — tool support was needed to work effectively with the JCF's score notation. The experiences from JCF analyses performed for the papers in this dissertation have provided extensive input for the development of the tool, and I have contributed to the development throughout the process, providing the user perspective on the tool to the development team. At the later stages, the tool has been of great help to perform the analyses. The first version of the tool was focused on the possibility to provide immersive demonstrations in virtual reality. The drawback of this approach was that the editing part of the tool become overly complex to build because of the chosen technologies and frameworks. Hence, a second version was developed with focus on editing the scores. A more modular architecture enabled the tool also to be used for recording joints directly, or as a scenario preparation environment with interfaced systems. Like for the first version, I contributed by extensive testing of the tool, and provided input for further development of the functionality and the user interface. During the later stages of my work, the tool had come close to a first stable version. When finalized, it will provide invaluable support for further studies using the JCF.

5 Summary of Papers

The dissertation is based on five of my papers, which are presented in the summary below. In addition, there are also three additional papers, Paper VI, VII, and VIII [58, 61, 62]. For all eight papers, except for Paper V (Section 5.5), I have been the first author and main responsible, and I have also been the presenter of the conference papers. For the Papers I–IV (Sections 5.1 to 5.4) this also means that I have been the main contributor to the whole paper, including the planning and execution of the practical work with the studies, and leading the activities. The same applies to the additional Paper VI, the doctoral consortium paper [58] where I, naturally, was the sole author. For the additional Papers VII *Untangling Complexity in Control Process Interactions* [62] and VIII *ATCO Strategies in Tactical TBO: A Simulator Study* [61], the writing was more of a collaborative art, even though I had the main responsibility for the papers. Technical development such as eye tracking integration or simulator adaptations has mainly been performed by LFV software engineers or research engineers from LiU. My contribution to the studies in the included papers are further detailed in the summary of each respective paper. Each main paper is described separately in Sections 5.1 to 5.5, while the supporting papers and other additional work are accounted for in Section 5.6.

5.1 Paper I

Attention Support With Soft Visual Cues (Nylin, M., Lundberg, J., Johansson, J.), 2020

The first paper [60] addresses communication from automation to human from an attention, and attention support, perspective. Current control room environments put a high visual load on the operators, and information can easily be missed. The main focus of the study was on ATC, but it also included a complementary part with Train Traffic Management (TTM) and Vessel Traffic Service (VTS). For ATC, a specific case where this might be a problem, the visualization of conflicts in the Conflict And Risk Display (CARD), was identified though operator workshops and incident reports. The visualization represents conflicts between aircraft detected by the Medium-Term Conflict Detection (MTCD) in a scatter plot visualization, where each conflict is represented by a square. However, if the conditions change, or new conflicts appear, the change may not be notified by the ATCO. If the change is to the worse, i.e., separation cannot be granted, it is important that the ATCO notices this, preferably before safety nets are triggered and last-minute recovery actions are needed. Hence, the ATCO should be notified if such information is missed. However, in an already information-rich environment, how can this be achieved without interrupting and adding even *more* cognitive strain on the ATCO, or drawing attention from something potentially more important of which the system is not aware? Current attention support types in the form of alarms, attention guidance, and subtle gaze direction were found not to fulfill these requirements, since they actively draw attention to the object or area of interest. The concept of *Soft Visual Cues* (SVC) was developed as complementary type of attention support. The underlying assumption is that the system, for example, by using eye tracking, can determine whether an object on the screen is looked at and thus likely noticed. If not so, it shall provide a gentle reminder, a Soft Visual Cue, to the operator that is recognized when the operator looks at the area around the object, but not drawing peripheral attention to the object of the operator looks somewhere else. Note that the SVC is intended for environments where the operators can be assumed to regularly scan the areas of interest as part of the work practice.

The aim of this paper was to evaluate different candidate design and determine which of them have the best potential to be successfully implemented

as a Soft Visual Cue, and furthermore, to get the operators' views of SVC as a concept. The main focus of the paper was on ATC and the CARD. Based on previous research on which types of changes in graphical objects are easy to recognize, I designed five different visualization candidates for modification of the CARD conflict symbols. Each design had a change of one property: color, opacity, size, rotation, and intersecting element, respectively. The designs were evaluated by ATCOs in individual workshops which I prepared and led. Each design was presented to them using a simple prototype, and they rated four statements about their first impression with respect to intrusiveness of the symbol itself, and of the property change. Furthermore, they rated to which extent the designs would fulfil the purpose of being a Soft Visual Cue in the intended environment. Finally, they got the opportunity to provide feedback on the concept as such.

In the complementary part with TTM and VTS, the operators performed similar assessments of elements in their respective graphical user interfaces. The train dispatchers evaluated sections on a digital rail plan, where the cues consisted of color change, slowly pulsating color, intersections, and size changes (line width). For the VTS domain, changes in vessel prediction lines in the Electronic Chart Display and Information System (ECDIS) were used. The properties changed were: color, change from full to dotted line, and size (line width).

The result of the workshops showed that the participants found the concept promising. The ATC SVC design that was considered most likely to fulfil the requirements of a Soft Visual Cue was a geometrical property change in the form of rotation. However, what type of change that would be most promising seemed to be domain-dependent, where results from the TTM and VTS domains indicated that more prominent cues were preferred compared to the ATC domain. This may be attributed both to differences in the control room layout and systems, as well to differences in process pace and control paradigm between the domains. The conclusion drawn was that SVC is a promising concept, but also that it requires a thorough analysis of the environment in which it shall be implemented since that will affect the design — a good SVC in one domain may be perceived as a shy whisper rather than a gentle reminder in another domain.

5.2 Paper II

Reduced Autonomy Workspace (RAW) — An Interaction Design Approach for Human-Automation Cooperation (Nylin, M., Johansson Westberg, J., Lundberg, J.), 2022

Albeit only one fifth of the included papers in numbers, the *Reduced Autonomy Workspace* (RAW) paper [59] is a major contribution to the dissertation as it articulates important principles for the F AUTO concept. The starting point for the paper was the fact that, despite good knowledge about issues that may follow increased automation, such as automation surprises and humans having to cope both with the work to be done and the automation that was supposed to help them, the same issues continue to appear. To solve this, I started to ask: why does it not work? Through studying relevant literature and having informal workshops and discussions with my supervisors as well as the F AUTO research group, communication between automation and human was identified as a critical point in the human-automation collaboration, especially when the automation initiates communication with the human operator. This happens when the automation, for any reason, needs to consult the human. This can be denoted as a reduced level of autonomy with respect to cognitive control during a certain period.

It was found that most frameworks that describe information exchange between systems and humans do not take temporal aspects of real-time processes into account. Rather, they describe a generic loop, or flow, of information exchange, or events. This may be correct in an abstract, conceptual way, but it does not provide much help for a designer to understand what is happening in and between different control processes. A similar problem was identified with the use of the notion of Levels of Automation (LOA). Like for the control process models, it might be correct to describe a system, or part of a system, as having automation properties and capabilities at a certain level of automation. However, the different notions, and there are several, mainly describe the division of tasks and responsibilities between automation and human within the levels, not how interaction should take place, neither within, nor between, levels.

This analysis resulted in the design approach called Reduced Autonomy Workspace (RAW). By using the Joint Control Framework (JCF), it describes how an automation should communicate successfully with a human coun-

terpart working, in parallel processes, with controlling the same objects, but at a higher Level of Autonomy in Cognitive Control (LACC). The JCF enables modelling of control processes over time, which is a key element in the RAW. By using sensors and other technologies to allow the automation to understand when communication is appropriate to initiate, the risk for inconvenient timing of communication initiation is reduced. Furthermore, the human must also have enough time to respond to the automation, while being informed of the respective amount of available time to be able to prioritize among tasks. Additionally, the automation shall be able to transform the information presented for the human, to a LACC that matches the level at which the human is working, such as setting and following up on goals in a control process. This is important — if the information is presented at a level that is far from that of which the human is working, it is cognitively demanding for the human to make the transformation. This means that a minimum of three processes must be modelled and considered: first, the process of the operator working with an external process, in this case the air traffic; second, the process of the automation working with the air traffic; and third, the process of the automation monitoring the operator to understand his or her situation.

By applying this approach when designing automation, some of the problems with human-automation collaboration related to communication could be avoided. The content of the information gets more predictable and requires less cognitive effort from the human, and the timing should provide a smoother information flow.

5.3 Paper III

Reminding Without Disturbing: A Soft Visual Cues Attention Guidance Design for an Air Traffic Control Conflict Detection Visualization (Nylin, M., Söderholm, G., Lundberg, J., Bång M.), 2023

This paper continues the work on Soft Visual Cues (SVC) by implementing them in the conflict window visualization, Conflict and Risk Display (CARD), of a simulated ATM system. The main question in the study was to explore how well the suggested design would fulfill the purposes of an SVC, that is attracting attention when looked close to, but not otherwise. I based the design on the results of Paper I (Section 5.1), a rotation of the sym-

bol that represents the conflict. In addition, we also explored other aspects, such as whether the design should be different depending on the Air Traffic Controller (ATCO) role (executive or planner controller), what time horizons would be feasible, and if there are other parts of the ATM system where they could imagine that Soft Visual Cues could be implemented.

By using a system with a novel technique for real-time image manipulation, the symbols were modified in the simulator's video output stream before displaying them on the screen. The system was developed, based on my requirements, by a team of research engineers at Linköping University, led by the second and last authors. By this approach no integration with the simulator system used was needed, which reduced the complexity of the setup, thus the risk for bugs and regression errors, and the amount of time spent for integration and testing.

The Soft Visual Cue design was tested in real-time human-in-the-loop simulations at the LFV research and innovation facilities at the Malmö Air Traffic Control Centre using the Narsim simulator. The simulator is a research simulator with realistic physical flight models for the aircraft. It is not based on any specific operational ATM system, but has been adopted to mimic the operational ATM system, Thales TopSky, used for area control at LFV. The traffic scenarios used were based on real traffic, and the sector in which the ATCOs worked was a real area control sector. Eye tracking glasses were used to provide information about whether the conflict symbols had been looked at for a certain amount of time. Since the results from Paper I (Section 5.1) were not conclusive regarding whether the change in orientation of the symbol should be immediate or a gradual rotation, both were tested. During one week, a total of 18 simulation runs with 9 ATCOs were performed. The main data collection consisted of debriefings with each ATCO. After each run and during the debriefing they also rated the concept with respect to intrusiveness and fit-for-purpose respectively, using the same method as in the first SVC study, Paper I (Section 5.1). In addition, data was recorded regarding rotations of the symbol and when the ATCOs looked at the conflict symbol. These recordings mainly aimed at complementing the ATCOs' statements on intrusiveness, i.e., whether the change of the symbol draws attention to it.

None of the ATCOs experienced the SVC as being intrusive, or that it drew attention away from other things. This was also supported by the recorded data on gaze and symbol manipulation. It was generally considered a good

reminder, though some participants thought that it was too soft. To some extent, this could probably by attributed to the fact that all objects on the screen were slightly smaller than in the real system due to different screen size and resolution. The feedback from the ATCOs confirmed that using SVC on the CARD was a good choice and targeted a well-known problem. It was also confirmed in the feedback that the simulated traffic scenarios were realistic and reflected a normal day at work very well. No preferences with respect to gradual or immediate rotation were reported and no differences in detection times could be identified. A consistent behavior of the SVC for both the executive and planner controller roles were preferred, even if they work on different time horizons.

The results from the ATCO feedback indicated that the SVC worked as intended with respect to being a soft cue, i.e., not attracting unwanted attention, which was also supported by the recorded data. The feedback from the ATCOs indicated that it also served as a good reminder.

5.4 Paper IV

Glyph Design for Communication Initiation in Real-Time Human-Automation Collaboration (Nylin, M., Lundberg, J., Bång M., Kucher, K.), 2023

In the fourth paper, a design for initiation of communication between automation and humans in a future, highly automated environment was evaluated. The basic principles were the same as for the Reduced Autonomy Workspace (RAW) concept: to provide information at the right Level of Cognitive Control (LACC) for the human, and with respect to temporal aspects in the form of timing of initiation and time for response. Furthermore, it should also provide information about the consequences, or effects, of a negative or left-out response. The idea was to create a basic design that could be adapted to ATC, VTS, and TTM respectively, even though the starting point was the ATC domain.

Based on the requirements above and that the visualizations had to be compact and easy to perceive, glyphs were considered a suitable information visualization concept. Glyphs provide a possibility to present multidimensional data in a very compact format. As the design was intended for use in a real-time system, the number of visualized dimensions was limited to

four to avoid too long perception times, which is a risk if using complex glyphs. I designed the glyph based on the identified information that should be presented in the glyph. The basic element of the glyph was a circle, where the color of the circle should indicate the type of information. Outer arcs depicted time left for response and elapsed time since the glyph appeared, respectively, and symbols in the middle of the glyph depicted the effect on traffic in case of a negative or left-out response. An example of the latter is that an aircraft would have to make a non-optimal change of heading. More detailed information should be available on demand by the operator, but that was out of scope for this study.

To evaluate the glyph design, I organized three separate one-day workshops: one with ATCOs, one with VTS operators, and one with train dispatchers. The glyph was introduced element by element, and the participants were given examples of situations where it could be used to provide input for the discussions. Glyph reading exercises were performed to assess how the glyph communicated temporal aspects. The participants then identified situations in brainstorming sessions where the visualization could be used. The results thereof were then used to discuss the proposed glyph design, including things like suggestions on relevant glyph types, needed information of the effect of negative responses, and maximum number of glyphs that should be present at the same time.

From the glyph reading exercise, it could be concluded that it was possible to read the arcs representing time left for response and elapsed time, respectively, with good precision, in this case meaning not more than one or two minutes of error in a glyph where one full circle represented 20 minutes. There were a few occasions where larger errors occurred. When analyzing them, it was apparent that the participants had mirrored the numbers, for example answering 1/4 of possible time was left for response instead of 3/4, or vice versa. This indicated that a full circle was probably not an optimal time visualization. The participants' feedback on the need for visualization of elapsed time was not conclusive and must be further researched. The results showed differences between the three domains with respect to how many glyphs would be convenient to have at the same time, and also, how the types of communication should be categorized. Finally, it also showed that the suggested feedback of negative or left-out response was considered less relevant, and instead, the effect of it, such as longer flown distance, would be more convenient information.

The proposed glyph design was a first example of how the RAW design principles can be used as a starting point for a design work, in this case for the visualization of communication initiation. It is important to remember, though, that the visualization was aimed at specific domains with highly skilled and trained operators/users. Designs for other domains with other properties and users may require other types of visualizations to support the RAW principles. Correspondingly, the glyph visualization may be used to support other concepts than the RAW.

5.5 Paper V

Modelling of Operator Control Work Across Traffic Management Domains: Implications for Interaction Design (Lundberg, J., Nylin, M., Praetorius, G., Arweström Jansson, A., Bång, M.), 2023

A key component in the F AUTO project has been the cross-domain approach. In this paper, this was in focus. We addressed the matter of similarities and differences between the domains from a control process perspective and how these can be modelled and analyzed. My main responsibility in this study was to manage the parts for the ATC domain, including the workshops held. For the other domains, I participated in the overall analysis and supported them in the JCF analysis work. I also contributed to the cross-domain analysis together with the first author.

In two workshops with domain experts, common scenarios from all three domains of ATC, VTS, and TTM were identified and analyzed. In the first workshop, the focus was on finding situations where humans interact with a system and where there are challenges in this interaction involved, for example, if the understanding of the situation differs between the human and the automation, or if information is not properly provided by the automation. The common situation across domains found was the situation of a late discovery of a deviation from the assumed progress in a vehicle process. In those situations, information has been available in the system for some time and presented to the operator, but have not been observed. What happens in this moment from a control process perspective, what are the differences between the domains, and can similar solutions be used across domains in the design of future automations?

As a preparation for the second workshop, specific scenarios were prepared based on real situations. For ATC, it was a late discovery of a conflict between two aircraft that was initially solved, but where conditions changed and a risk for loss of separation arose. In this case the situation was first discovered by an automated system for conflict detection, but not seen by the controller. In the VTS case, a vessel has missed to perform a turn and the VTS operator discovers the situation late. Finally, in the TTM case, a train dispatcher misses that a signal is still set to red/stop when it should have been set to green/go. The situation is not discovered until the dispatcher receives a telephone call from the train driver, who asks why they are not allowed to continue their travel.

During the second workshop, another set of operators than in the first study from respective domain worked together with the researchers to make an initial Joint Control Framework (JCF) analysis of the situation using a physical JCF score at a white board, where post-it notes represented different joints. The analyses were then transformed into digital format using the JCF editor tool. Different processes and episodes were identified from the original single, overview score format. Finally, the scores were refined by method experts so that the situations from all three domains could be compared.

The comparison focused on what happened when focus shifted from monitoring processes to the processes-in-focus of the critical situations and concluded that this shift is a core problem for all three domains. The results revealed differences in how the shift happened and especially how the processes-in-focus looked like, while monitoring the overall situation was more similar. There were also differences in the interactions that took place between discovery of the deviation till the situation was solved. The ATC and TTM processes included relatively few cognitive joints in short episodes, while in VTS, there were several joints, not least of information perception and interchange to build up an awareness of the situation. TTM was found to be more pro-active due to the built-in control in the system, while ATC and VTS were more reactive. This indicates differences in the framing of the processes' focus between the different domains. Overall, though, the domains were quite similar from a joint control framework perspective. For future design of human-automation interaction, F AUTO principles were concluded to be one possible way forward. For helping operators attend missed situations, it was concluded that only increasing salience is not sufficient. More elaborated solutions are needed to avoid adverse side effects. For a larger

paradigm shift with a fully autonomous automation, the automation must fully understand all processes and how they affect each other.

Summary of the Paper Conclusions

The question relating to how flexible automation shall be designed on a conceptual level (**SRQ2**) is mainly addressed in Paper II (Section 5.2) and Paper V (Section 5.5). These papers also include discussions related to **SRQ1** about possible underlying problems. The main conclusions from the RAW design principles in Paper II (Section 5.2) were:

- Initiate communication when the operator is available for communication.
- Make sure that there is time for response.
- Let the automation transform the information to a LACC that matches the operator's, thus avoiding cognitively demanding changes of LACC by the operator.

In Paper V (Section 5.5), the main contribution was that it shows how different domains may share properties, but also differ, when focus is shifted between general, monitoring control processes to specific processes. Similar to Paper II (Section 5.2), it highlights the importance of the temporal aspects in the control processes. Possible ways forward in the design of highly automated systems were identified: either in the form of complete responsibility for monitoring being assigned to an automation (which would be a paradigm shift), or, in the use of F AUTO principle as long as the automation is limited.

How to design interactive visualization to support flexible automation (**SRQ3**) was addressed in Papers I, III, and IV (Sections 5.1, 5.3, and 5.4, respectively). The first two papers dealt with the Soft Visual Cues concept. The SVC focuses on environments where full flexibility is still limited and the only input to the F AUTO system is the operator's attention. The SVC provides a complementing concept of attention support that:

- Reduces the risk for unintentionally catching attention in unsuitable moments.
- Has been tested in real-time human-in-the-loop simulations.

While not focusing on **SRQ3**, Paper V (Section 5.5) concludes that only increasing salience or adding alarms may be a less good idea unless addressing problems of framing the situations at the same time. If not addressing the framing, it may disturb the operator in his or her ordinary traffic scan, which may in turn affect situation awareness and interfere with the operator's plans, thus highlighting the need for supportive visualization concepts.

Paper IV (Section 5.4) shows how visualizations for initiation of communication, built on the RAW design principles, can be designed by combining current knowledge of information visualization with the RAW design principles. It also provides a cross-domain perspective where the basic principles were shown to be possible to visualize in the same way in the included domains. At the same time, it identifies types of information that are domain-specific, thus implying that a good domain knowledge is crucial when designing the visualization.

5.6 Additional Work

In addition to the main papers upon which the dissertation is based, the work resulted in additional publications in the form of papers and posters. The additional papers are briefly described in subsections below. When still planning the main project, I participated in the analysis of an eye tracking study of missed events in an ATC tower environment. The study showed how easily information in an information-rich environment can be missed, and it pointed at the lack of automated support to mitigate this risk [49]. The same year, 2016, I participated in the Single European Sky ATM Research (SESAR) Innovation Days (SID) with the poster *A New Modality for Air Traffic Control* about how sonification could be used to decrease visual load on ATCOs. At the SID 2017 I participated with two posters.[1] The first poster described the upcoming F AUTO project. The second poster, *Back to Basics*, was based on the observation that complexity seems to increase in the ATM functional systems, almost as a law of nature. The idea presented was to start from a blank paper, instead of tweaking current concepts, as a way of generating new ideas without being restricted by current systems and paradigms.

[1] Note that posters from SID earlier than 2019 are no longer available on the SID website.

My research has also been communicated in other fora, complementing the academic publications. For example, I have participated with presentations of my research in different contexts, such as the Swedish bi-annual conference on transportation research, Transportforum, held by the Swedish Road and Transport Research Administration, at the Swedish Human Factors Network (HFN) workshops, and at internal seminars, both at the Linköping University and at LFV. Below follows a short summary of the additional papers produced during my PhD period.

Paper VI

The first additional paper was a short doctoral consortium paper at the 31st European Conference on Cognitive Ergonomics conference (ECCE2019) [58], where I presented my planned research, including what was to become the SVC concept. This contribution was complemented by a poster on the same theme at the doctoral poster session.

Paper VII

The paper *Untangling Complexity in Control Process Interactions* was a position paper for the 2021 ACM CHI Virtual Conference on Human Factors in Computing Systems (CHI2021) Emergent Interaction Workshop [62]. It addressed the problem of analyzing large amounts of data from complex, entangled control processes, such as in Air Traffic Control. Manual, qualitative analyses may provide details on the interaction on a certain occasion, but it is hard to find patterns in the overall picture as the analyses are time demanding. Quantitative analysis methods such as clustering are useful to find patterns in the data, but the results lack context. What do the patterns mean? We suggested ways of combining these different methods to get both contextual understanding of patterns, and to find patterns among multiple entangled episodes analyzed in detail. It was also hypothesized that future development in AI will open new opportunities for analyses of complex, entangled processes that have not been available so far. The idea was presented and gained quite some interest at the workshop. It has not resulted in further studies in that direction, but I nevertheless consider it a very interesting and relevant area to explore. It can be foreseen as important for future work on flexible automation, especially if it can be implemented into real-time applications, and not only used for analysis of recorded data.

Paper VIII

A project that emerged from the work during a course in cognitive systems engineering resulted in a study about how the introduction of a tool for tactical Trajectory Based Operations (TBO) in an ATM system would affect the ATCO strategies for solving conflicts between aircraft. The tool was implemented in the Narsim simulator and tested in a small scale real-time human-in-the-loop simulation [61]. The paper was presented at the 33rd Congress of the Aeronautical Council of the Aeronautical Sciences (ICAS), in 2022.

The new tool allowed the ATCOs to directly manipulate complete trajectories and send the new trajectories to the aircraft, which, if used, reduced the need for monitoring the conflict solutions compared to when using radar vectoring for lateral solutions. The tool allowed for vertical solutions as well, but that was not used by the ATCOs. The overall principle of the tool was positively received by the ATCOs, even though the implementation of the tool was considered possible to improve a bit to reduce the number of inputs. The study also highlighted the differences of strategies used by ATCOs regardless of tools used (results from a larger study using the same setup and traffic scenarios were used for comparison), which is an important aspect to consider in all interaction design. While not directly related to flexible automation, it was relevant for how increased automation can affect human-automation interaction and how the new automation affected the ATCO traffic conflict solution strategies.

6 Discussion

The overall objective of this dissertation was to explore how human-automation collaboration can be enhanced by the design of flexible automation and associated visualizations. One of the first things I did when I started my work, was to perform an overview of the current state of the art in adaptive automation research. It became apparent that there were still many issues left to solve, many of which were related to automation in general, not only with respect to adaptivity. The problem space seemed to be relatively well explored and described. When beginning to scope the problem and working on the first drafts of concepts, I realized that there was a genuine risk of ending up with the same, already known problems myself. As a result of this, I chose to use the existing knowledge about the problems and issues as a starting point for my further work in exploring concepts and visualizations for flexible automation. This resulted in two main contributions: The concept of *Soft Visual Cues* (SVC) as a complementary new form of attention support, and the *Reduced Autonomy Workspace* (RAW) which is a design approach for initiation of communication between an automation and a human operator. The latter was also accompanied by a glyph-based visualization concept.

The results from the studies have already been described in the Summary of Papers, Chapter 5. In the following sections, the results from the papers

and insights gained throughout the work are discussed and reflected upon, following the main steps in flexible automation: sensing, interpreting, and adapting. After that, some reflections on the work and research process follow. Finally, I present an outlook on flexible and increased automation in a wider context.

6.1 Sensing the Operator

How to measure the cognitive state of the operator, for example, to assess if the operator is busy, is important for the concept of flexible automation, but it has not been a core topic for my work. Rather, I have assumed that the technology needed will be available and function as intended. Nevertheless, I have used sensor technology in some of my studies and I have had reason to reflect on the subject, based on my own experiences. For those cases where I have used sensors, it has been in the form of eye tracking, both for live attention measures by gaze fixations, and collection of gaze data for post-analysis. In the F AUTO project, it was established early that sensors must be non-intrusive. At the same time, we were aware that many sensors currently available are far from non-intrusive, which was also confirmed by the initial literature review on the adaptive automation. In the SVC simulator study (Section 5.3), eye tracking glasses were used. Most participants did not have any problems with them, but there were situations where they were considered (too) uncomfortable, which is in line with previous experiences. Furthermore, in a live environment, it is a must that it can be used together with glasses for correction of vision, or that the eye tracking glasses have built-in correction. This was also a limiting factor when recruiting participants. Camera-based eye trackers like the one used for the study in Paper VIII (Section 5.6) are less intrusive, but instead, they introduce other challenges, such as what happens if two people are sitting in front of the screen or if one is moving around too much.

The general impression of both the camera-based eye tracker and the glasses was that the precision and the reliability of precision over time needs to be improved if this approach shall be used in an operational environment. There were no concerns raised during the studies about the recording of eye gaze data, which could presumably be a sensible topic. In these studies, the data was only used for research with people participating voluntarily (and as part of their employment), and it was only a limited capability of the eye

trackers that was utilized (*x,y* gaze coordinates). However, much more information can be retrieved from modern eye trackers, which can be used to get information about, for example, the medical condition of the operator. Furthermore, future development in analysis methods may make it possible to reveal more from the data than what is possible today. It could also be sensitive if recorded eye tracking data is used in incident investigations, even though it could as well be argued that it would be a help and security for the operator. As mentioned, no such concerns were raised during the simulations. Nevertheless, it is a perspective that must be considered when using *any* kind of sensors on the operator — keep the sensing to a minimum of what is needed, and be transparent about what and why. Likewise, it must always be questioned what data will be stored and why, and for whom it shall be accessible.

6.2 Interpreting the Operator

The next step in the F AUTO process is to transform the recorded data about the human operator into an actual understanding (for the automation) of the operator's situation. There are several ways of measuring parameters related to workload and attention which can be used to assess the situation of the human. This information must then be combined with the knowledge of the included control processes and overall situation. What has become apparent during my work is that despite available technology, the challenge how to *interpret* the recorded data remains. For example: if the operator has looked at an object, what does it mean? Even if the object has been seen, how has he or she processed the information? Did it cause any action, or did it affect any decisions? And if it has not been seen, is that because the operator has identified the situation in another way, or is the operator occupied with something that is even more urgent? The SVC concept was developed as a bridging step — if a certain object has not at all been looked at, the information has not reached the recipient, and a gentle reminder is introduced that shall not disturb the operator. Understanding why the information was missed is not needed. However, this is a passive form of adaptation and a situation of low complexity; the information is either looked at or not looked at. If the automation shall have a more active role and be able to judge whether a soft visual cue or a more salient form of communication, such as attention guidance, or even an alarm, is appropriate, then the system must have a more

elaborate understanding of the operator as well as of the overall situation including the controlled traffic processes. Possible ways forward from this perspective were discussed in Paper V (Section 5.5).

Attention and Intention

Looking specifically at attention, an operator may deliberately disregard some information for some time to focus on what is most important for his or her plan to solve the tasks at hand. If the automation could understand what those plans are, the automation would then be able to understand if the ignorance of the information is on purpose or not. However, this is a very complex task. First, the controlled processes are hard to predict. As discussed in the additional paper on entangled control process episodes, Paper VII (Section 5.6), continuous small changes in the conditions make seemingly similar situations evolve differently, and different processes affect each other. Second, human decision making is not necessarily a strictly logical and consistent process where we evaluate all possible solutions and then pick the best one. As pointed out in Section 2.3, this can very easily be observed in real-time simulation studies such as our study on the trajectory-based operations, Paper VII (Section 5.6), or the Soft Visual Cues simulation study, Paper III (Section 5.3). The simulation scenarios start with exactly the same conditions, but the ATCOs choose different solutions following different plans. The result is that there are not two identical scenarios, and that is in a controlled environment, where the sources of differences are the ATCOs' actions and the timing of the implementation of the solutions by the pseudo pilots handling the simulated aircraft. This applies even if the same controller runs the same scenario multiple times. Hence, if a highly automated system shall be able to provide active help to the operator, it is likely that it must understand the intentions of the operator, or at least be able to some extent assess the probability of different choices and actions. In a way, it can be seen as an equivalent to how automation transparency is supposed to create an understanding of the automation for the human. Following this line of reasoning, research on automations' ability to interpret the humans' intentions should be included in future F AUTO research. This could, for example, be combined with the principles in the RAW design approach to identify *when* to communicate.

How Busy is the Operator?

The attention of the operator relates to how busy the operator is, which in turn is important for the timing of communication initiation in the RAW design approach. In the work with the RAW, I did not specifically define *how* it should be determined by the automation if the operator is available for communication. Two ways of measuring this are by estimating either task load or workload, and there are good reasons to investigate this in future research. Task load has the advantage of being easier to identify and quantify. The disadvantage is that it only gives a hint about the effect on the operator's workload. It requires knowledge about how the tasks affect the operator, which may vary both between individuals and within individuals over time depending on their status and other external factors, not directly related to the task. Hence, the same task load can lead to different workload. To make it even more complicated, it must also be related to how the workload impacts performance. All of this must be considered when designing flexible automation and is an important area for future research on flexible automation. It could also be an interesting topic for future AI research — can new technology help us to identify and use patterns of task load, workload, and performance in real-time to enhance the understanding of, and predict, the operator's situation to achieve flexible adaptation? Once the interpretation of the operator's situation is solved, it can be assumed that it becomes easier to design well-functioning adaptation that is fully flexible in the F AUTO sense.

6.3 Adapting to the Operator

Adapting the information presented to the operator was addressed both in the RAW design approach and in the SVC concept, which included conceptual ideas as well as specific visualizations to support the ideas.

Flexible Automation and Levels of Automation

In the RAW paper, see Section 5.2, the shortcomings of using levels of automation as (the only) basis for flexible automation were brought up. The main issue addressed was that it does not describe the interaction between the human and the automation at the different levels, or what happens when

levels of automation shift. Hence, using levels of automation is appropriate when describing how tasks and responsibilities are, or shall be, split between the human and the automation. It is also a convenient way to communicate around to what extent a system is automated, even if it can be discussed if it is more relevant to talk about complete systems or about subsystems. However, the latter is a discourse that is not directly related to flexible automation. Even if automation level taxonomies may not be convenient to use on their own to form the basis for flexible automation, they can be used together with the notion of Levels of Autonomy in Cognitive Control (LACC), which was described in Paper II (Section 5.2) in a LACC-LOA matrix. By doing so, both who does what, and how the collaboration is carried out can be brought together. The point here is that even if the different LOA taxonomies have their shortcomings from an F AUTO perspective, they can still be relevant.

Situation Awareness and Reduced Autonomy Workspace

None of the studies explicitly addressed Situation Awareness (SA) as a main focus area. However, loss of SA is a consequence of many of the problems and issues in human-automation collaboration. Furthermore, it was established in Paper V (Section 5.5) that the JCF analysis implicitly becomes an analysis of the SA. Consequently, even if the RAW aimed at providing a design approach to avoid problems *leading to* loss of SA, it may also be used to illustrate the problems from a loss of SA perspective. At any point in time, the focus of a human operator will be at something, at a certain LACC, and within a certain time span. The time span basically starts at the current point in time, and stretches forward in the forms of plans made for future actions and decisions to be made. To some extent it may also include reflecting on what has recently happened. Within these limits, the operator maintains an awareness of the situation. If the operator is working on a high LACC within a process, and the automation is working in parallel, but at a low level, both can have an understanding of the situation with respect to, for example, the control of an aircraft, but at different LACC levels (to avoid misunderstandings — I do not equal the automations' so-called understanding of the situation with the human's situation awareness). See Fig. 6.1 for an illustration. However, if communication is initiated by the automation outside the human's boundary of awareness, confusion and loss of SA may arise. It could

probably also lead to automation surprises. Hence, this could be another way of looking at the RAW approach, and to have in mind when designing new automation; the automation shall be designed to initiate communication within the operator's awareness with respect to LACC and temporal extent, which includes the understanding of the immediate plans and intentions of the operator. Even if the RAW principles are followed and the automation is designed to be able to adapt its communication, there will certainly still be situations where it is not able to keep the communication within the awareness limits of the operator. If so, the operator must be given both time and supportive information to reduce the cognitive effort needed to change the focus to be aware of the new situation. If that is not possible either, the automation must be able to sort the situation out by itself, or, by layers of other back-up/help systems.

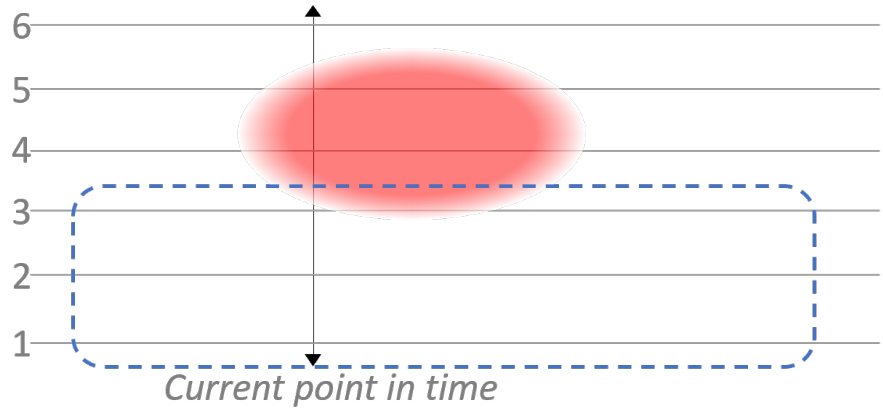


Figure 6.1: The red oval shows within what limits the human maintains awareness of the situation and the blue rounded-corner rectangle where the automation is working, all presented in one common, simple JCF score. If communication is initiated about something outside the awareness of the operator, he or she will not be aware of it and have to spend cognitive effort to create awareness, thus risking to move focus and awareness from other things. Hence, the automation should be able to understand within what limits the human maintains his or her awareness of the situation. This is another way to illustrate why adaptation of communication initialization in accordance with the RAW principles is important.

Shifting Between Control Processes

One of the identified issues with automation is that the operator needs to monitor and understand what the automation is doing to be prepared when the automation must consult the human, or when the automation fails. The RAW approach should reduce the need for this in nominal cases. This should

enable the operator to focus at his or her own control processes, being confident that they do not need to monitor the automation. However, the RAW does not, and was never intended to, address automation failure or shortcomings of the automation. Note that if the automation reaches the limit for its autonomy, the autonomy is reduced, but it is not a failure. However, if the automation is not designed to handle its reduced autonomy, for example, by consulting the human, it may result in an automation failure. My standpoint is that systems must be designed for graceful degradation, *not* leaving the human as a backup. Having that said, a possible development of the RAW ideas could be to explore if it can be used to design specifically for communication during degradation. Can the same principles be applied to let the human operator take care of the failure at a convenient LACC and at a convenient point in time? If so, it could help to reduce the need for monitoring the automation also with respect to possible failures.

A similar problem was addressed in Paper V (Section 5.5) when an operator must shift between working with larger, general monitoring processes to control processes relating to a specific object in a specific situation, processes–in–focus, for example, two aircraft in conflict. A flexible automation that identifies the situation to be in focus is one way forward suggested in the paper. Another possible way forward presented was to let the automation take over the monitoring tasks completely. However, this would be a paradigm shift that would require much more competent automations than available today. However, in both cases, it is likely that communication with the operator is still needed at some point, and then RAW principles should be applicable. This could be especially important when the automation has identified the situation to be in focus and need to consult a human operator to solve it.

6.4 Reflections on the Work and Research Process

The work carried out that resulted in this dissertation has included elements that I consider need some extra reflection. This includes the cross-domain research approach, my own background as an industrial PhD student with an extensive experience both from the domain and from the organization, and the experience of performing research related to a small community.

The Cross-Domain Approach

As already mentioned, the cross-domain approach of the F AUTO project has been an important aspect, and even when I have focused on ATC, the VTS and TTM domains have been present in many discussions, and in my own process of thought. The strength of the cross-domain approach is that it introduces more perspectives, and ideas and concepts can be tested and thought through more thoroughly. I have performed my work with a grounded knowledge within one of the three domains, which has been an advantage with regard to understanding the other domains, and with regard to how the domains relate to each other. Without such experience from one of the domains as a starting point, or foundation, I am quite certain that I would have needed to have a slightly different approach to my work to avoid spending too much time just trying to understand enough of the domains. Regardless of whether one has the required domain knowledge, or if that is achieved by teaming up with someone who has, I would, based on my experiences, recommend concentrating on one main domain. I do not consider that my work would have been possible to perform if I had split my focus equally between the three included domains.

Research Within the Organization

I have approximately 20 years of experience from LFV and the ATC domain and my educational background includes an area control ATCO education, even though my own operational experience is limited to eight months of on-the-job training in a (now) legacy system. This has been an invaluable asset in my research, which has allowed me to focus on principles and ideas related to my core research area of flexible automation without having to spend too much time trying to understand the domain. Likewise, things such as the procedures for requesting resources for my studies are well-known to me. Furthermore, my offices have been situated in the Swedish Air Traffic Control centers, so I have had a very direct connection to the daily work (with the exception for the pandemic years, when I was forced to work from my home office most of the time). Hence, I have a grounded knowledge about the domain, both from a theoretical perspective and from a practical perspective. This has probably allowed me to get into working with actual concepts and to perform studies on my own quicker than would have been possible without this experience. However, at the same time, I am aware that there

may be risks with this deep domain involvement. For example, it is is easy to assume that how Air Traffic Control is performed today is more or less axiomatic — "we do like we do because that is how we do things", so to speak. To counteract being stuck inside my own domain bubble, I have had great help of co-authors and colleagues in the human-automation interaction group at the university, as well as in the F AUTO research group. The transversal approach of the F AUTO project has also been an asset in this respect by providing new perspectives from similar, but not identical, environments and situations. A final reflection on this topic is that after two decades in the organization, I know a lot of people. Therefore, an important principle has been to recruit participants in my studies via standard company resource management procedures, rather than via personal connections.

Participants From Small Communities

The issues of performing research within a domain which is relatively small was touched upon in the chapter about operator involvement, Section 4.3. In addition to what has already been mentioned, there are a few other reflections. A limitation not directly related to the small community, but an effect of that in combination with the nature of the dynamics of the whole aviation sector (traffic fluctuations over the year) and working time legislation (mainly relating to summer vacations), is that the time of the year when it is possible at all to get participants is limited. Often, there is a period in the late spring and autumn, respectively, when it is possible to get participants for activities such as a simulation.

The problem with small communities of specialists is not in any way specific to the ATC domain (or for the VTS or TTM domains), and there are aspects that are more general. To minimize the participants needed over time, one strategy has been to always start in small scale to find out if, for example, a new concept is feasible at all, or if it can be rejected, or refined, at an early stage without involving too many persons. The limited number of available participants can also affect the choice of methods. Even if it would be preferable to perform a large-scale experiment to gather quantitative data, it might simply not be possible. Mixed method approaches may be a good alternative to counter this problem, where complementary quantitative data can be collected even if the emphasis is on qualitative data, such as the ATCO's experience of a new concept. In that case, having a relatively homogeneous

group, as was the case with the ATCOs, is an advantage with respect to representativeness. Another aspect is that in a small community, it is more likely that the participants will meet and discuss with each other about their experiences of their participation in a study. If they participate serially, there is a risk that they affect each other's opinions. It may not always be a problem, but if it is, it might be better to consider, for example, workshops where they all get the information at the same time and perform individual as well as cooperative exercises during the workshop.

Real-Time Human-in-the-Loop Simulations

To achieve full external domain validity (or ecological validity [32]) in studies of new ATC concepts, the best would probably be to test new concepts and designs in a real, live environment. However, that is most often not possible in domains like ATC, mainly due to safety reasons. This is why real-time human-in-the-loop simulations are common within the ATC domain. It is a technique extensively used in the education of ATCOs as well as for re-current contingency training, and it is also often used for research and development. Because of this, the ATCOs are used to work in simulators and to perform the work as they do in in the live environment. However, simulators are by definition a simplification of the real world. The look and feel may be similar, but not identical, and the available functionality is often limited compared to a complete live system. Furthermore, even if acting realistically in the simulator, all actors involved know that it is not the reality. The experience from the simulations performed for this dissertation (Papers III and VIII), as well as from other simulations, is that there are individual differences in how participants handle dissimilarities and limitations. Some just accept the differences, while others find them more disturbing.

The Narsim simulator used in the real-time simulations I have performed is not a live ATM system, but purpose-built to mimic real ATM systems for research activities. The fidelity of a simulator is always a trade-off between realism and complexity. On the one hand, too high complexity increases the risk for technical errors and failures, and increases the resources needed to perform a simulation. On the other hand, it must be realistic enough to allow the participants to evaluate the simulated concept/technology. The same is also true for the simulation setup, for example, with respect to surrounding sectors and traffic. A complex setup may also make assessment of

the results more difficult — what affected what? However, a less complex setup may make the ATC task easier than in real situations. In Narsim, this has been managed by working closely with ATCOs to include the most important tools and functionality. This involvement of domain experts in the simulator development is important to maintain (and improve) quality, and something that should be prioritized in any similar context.

6.5 Increased and Flexible Automation in a Wider Context

There is no doubt that increasing automation, not least in combination with AI, will affect how we work (although it is not always easy to predict *how* in the long run, see, e.g., [1]). On the one hand, it helps people perform in a way that would not have been possible without the automation. For instance, the amount of en route air traffic handled today would be impossible to handle without the current system support, even though the automation in current ATM systems is still on a relatively low level. On the other hand, it can also be perceived as a threat against one's professional skills, or even as a threat against the role as such — will I be replaced by a robot or automation? During my work in F AUTO, and from other personal experience from working with research and development, I have noticed that it is often the unknown that scares people. Current automation is familiar, and when pointed out, people may not even realize that the tools in the ATM system that they use in everyday situations are in fact automation. This is supported by Svensson and Rönnberg [90] who showed that that trusting and understanding the automation are important for ATCOs. It shall also be kept in mind that people may have experiences of less well-designed automations. The point here is that it matters not only how we design automation, but also how we talk about it, and how it and presented and introduced. Having that said, increasing productivity *is* a driving force behind automation in ATC, and even if the goal is increased output from the human and automation together, the effect (at least in the long run) will be that fewer people are needed to handle the same amount of traffic.

The F AUTO project has included control room environments from three domains, but there are other domains where similar control room environments can be found, such as control rooms for traffic management in metros or for

surveillance of road traffic. The emerging use of drones will probably create new types of control rooms. The sheer number of drones in the air makes traditional control of single entities, like aircraft in ATC, impossible. This drives development into new solutions, and new roles will emerge. However, even if the drone control systems need to be more highly automated, there will still be humans in the loop somewhere that need to interact with the automation. This is also an insight that is applicable on a more general level of human-automation collaboration. Even if automation is introduced to perform work previously performed by a human, there will always be interfaces between humans and the automation at *some* level. With humans and automation working together, but on different levels of automation (or LACC), well-functioning human-automation communication and collaboration can be assumed to be increasingly important. It is also reasonable to assume that the issues and suggested solutions and principles described here will apply to other domains and environments.

The automation addressed in the included papers mostly relates to pre-artificial intelligence (AI) automation, even if, for example, Paper V (Section 5.5) mentions human-AI/automation teaming in future systems. However, the new possibilities opened by the AI development during these years probably have the capacity to alter how we look at automation and human-automation collaboration. Which domains and areas will be affected and in what ways is very hard to predict, and I will not even try to do so. I believe, though, that we have just seen the beginning of what is to come from the AI technologies. Hopefully, it can possess unforeseen possibilities for flexible automation, however, care must be taken. Like pointed out by Endsley [21], relating to the classical automation ironies, we may encounter new challenges which we must identify and be aware of, so that we do not just solve one problem and then create new ones in the same process.

6.6 Reflections on Validity and Ethics

The above sections include reflections on validity aspects of my research as well as ethical aspects. The following paragraphs present these aspects in a more concentrated form. The research has targeted three domains (ATC, VTS, and TTM), and ecological validity [32] has been essential. Including operators from these domains has been the main way of addressing this. Ecological validity has also been addressed by performing studies in en-

vironments as realistic as possible, for example, by using traffic scenarios based on real traffic in a simulation, as in Papers III (Section 5.3) and VIII (Section 5.6), or to let the *participants* identify relevant situations to discuss as in Papers IV and V (Sections 5.4 and 5.5, respectively).

Due to the aforementioned difficulties of small communities with limited resources and access to operators, other studies have sometimes been performed with the participants other than domain experts/operators. This could, for example, include operators under training (students), retired operators, or engineers that have got a brief training for a specific task to be able to participate in the study. This increases the number of participants in, for example, a simulation. However, it also means that the results will be less related to the actual, day-to-day work, and in the example of students, they lack experience which may be needed, for instance, when evaluating a new concept. Based on this, operators with up-to-date operational experience were prioritized in my studies, even if that meant fewer participants.

Few participants may affect internal validity if a participant is in some sense extreme, and just a few persons dropping out may have large effects on the study, even if it is a problem mainly affecting quantitative methods [14]. As already discussed, having participants with the same training who were once recruited as operators by the same requirements should reduce the risk for the former. Furthermore, the participants were recruited to the studies mainly based on availability, which gives some level of randomization. The only other requirement (with the exception of not using eye-sight correctional glasses when eye tracking was used) was that they should be active operators. Regarding the risk of participant drop-out, it was a risk that could not be avoided. Seen from a qualitative research method perspective [14], examples of positive aspects regarding internal validity are my own profound ATC domain knowledge, presentation and discussion about discrepant information, and the reflections on my own possible bias in this dissertation.

The aspect of external validity with respect to generalizability to other domains is difficult to assess. The comparison between the three domains in F AUTO was addressed in Paper V (Section 5.5), and it gives some directions for how comparisons can be made with other, similar domains. It is, however, hard to draw any far-reaching conclusions for domains other than the studied ones without first making such comparisons, even if there is nothing to point against applicability of the results in other domains. One advantage

for the validity between the studied domains has been the use of the JCF notation, which has helped to keep a consistent coding of modelled control process interactions.

Working with automation may raise ethical concerns, since automation is often developed to replace human labor. However, technical development and automation have contributed to the ultra-safe aviation sector of today, and humans are still an essential part of the ATM business. Nevertheless, people's concerns about increased automation must be listened to and taken seriously. I consider viewing automation as a help to improve our capabilities is a good starting point, both for developing well-functioning automation, and to reduce fear of increased automation. The climate- and environmental challenges we face can raise concerns about increasing capacity to handle more flights. Increased efficiency creates more capacity, but it also increases the possibilities to reduce delays and optimize aircraft trajectories, which reduce energy consumption per flight. If it turns out that we need to fly less, flying as optimal as possible is still desirable.

Ethical concerns related to my studies, and also to future F AUTO technology, include the collection of data about the operators. Sensor data from eye trackers and other types of sensors, such as EEG and skin conductance, can be used to retrieve extensive data regarding the operator's medical condition, and advances within the field can be foreseen to increase the extent of this. In future F AUTO concepts, this is a question that must be addressed — it must be decided upon which data is needed for the F AUTO purposes, it must be transparent to operators, and it should be questioned if the data must be possible to access (and by whom) or if it can be used inside a closed system. No concerns were raised during the simulations where eye tracking was used, but it is reasonable to assume that the situation may be different if one is tracked on a daily basis by the employer, not just participating in a single experiment where they can cancel their participation at any time. Furthermore, the only data used in the simulations were gaze coordinates (not, for example, blink rates, pupil size, etc.). Similar to the concerns regarding automation, concerns about ethical concerns related to operator tracking must be listened to and seriously considered when developing F AUTO technology. Operator recordings could also raise more general questions about supervision of employees. What could, for instance, be the consequences if operators are being evaluated on performance using F AUTO sensors, or if the collected data is used for demanding accountability in case of an acci-

dent? Already today, a lot of what is going on in control rooms as well as aircraft cockpits, for instance, is recorded. This recorded data allows us to investigate things that go wrong, as well as to understand why things go right as well. However, it relies on the existence of trust and a just culture, which must be encouraged and developed.

7 Conclusions

To enhance human-automation collaboration, it is necessary to understand why the introduction of new automation often comes with problems and issues that are already known, which was one of the early insights in my work on flexible automation. During the initial phase of my work, I also identified that there was a lack of design approaches, or frameworks, to guide development of automation towards being flexible. Further on, it became apparent that the design of communication between automations and human operators is crucial to achieve the intended flexible automation. This is especially important when it is the automation that initiates communication with the human. The focus of the dissertation has been on communication through the visual modality, which was considered most relevant for the targeted Air Traffic Control (ATC) room environments. As presented in Section 1.3, this resulted in the overarching research question (**RQ**):

How can flexible human-automation collaboration be designed to avoid classical automation problems?

From the overarching, main research question (**RQ**), three sub research questions were formulated:

- *What causes already known issues and problems in human-automation collaboration continue to appear?* (**SRQ1**)
- *How can a design approach for flexible automation be formulated that mitigates known automation issues within human-automation interaction?* (**SRQ2**)
- *How can interactive visualization concepts support flexible automation?* (**SRQ3**)

7.1 Automation Problems and Design Approaches

Focusing on the first sub research question (**SRQ1**), it was identified that there are several notions that describe levels of automation, but they focus mainly on the division of tasks between human and automation and mention little or nothing about *how* humans and automation should interact, neither within nor between different levels of automation. It was also identified that temporal aspects are crucial for human-automation collaboration. Yet, most frameworks that describe human-automation collaboration do it without addressing temporal aspects in other ways than conceptually. Furthermore, these often use circular visualizations that describe, for example, perception, decision, action, and evaluation, but do not depict processes going on over time. The *Reduced Autonomy Workspace* (RAW), Paper II (Section 5.2), addressed these issues and presented a design approach for human-automation collaboration, thus addressing **SRQ2**. Special focus was on communication between a highly autonomous automation and a human when initiated by the automation due to a need to consult the human. Key aspects in the RAW are:

- Timing of the initialization of communication with respect to the human operator's situation.
- Adaptation of the content in the communication to match the Level of Autonomy in Cognitive Control (LACC) at which the human is working.

- Provision of enough time for evaluation and feedback from the human.

The RAW contributes to a better understanding of how human's and automation's control processes relate to each other, and how the understanding thereof can support the design of human-automation collaboration. At a minimum, three control processes must be considered:

- The control process with the human as subject.
- The control process of the automation as subject controlling the same object as the human.
- The process of the automation monitoring the human to be able to perform the needed adaptations in communication.

The last process, the automation monitoring the human, is the key for achieving flexible automation. By using the Joint Control Framework (JCF) score notation system in the RAW, it was possible to visualize the adaptation both in LACC and in time for use in design of new automations. It was suggested to adapt temporal aspects in real time, while the adaptation of LACC levels should be made off-line. The rationale behind this was that it is easy to state that the human operator shall only work at high LACC, but it was much harder to define it in a way that can be used by an automation. This requires more research, but if the adaptation of LACC by the automation can be solved, it would reduce the need for the human to understand the automation, thus solving the problems with the ironically added need for knowledge about the automation. Combined with the temporal adaptation, it should reduce the risk for surprises or the risk that the operator is forced into a situation with unnecessary time-pressure for making decisions. Even a predefined LACC should reduce the risk for surprises, however, it would be less flexible. So, the RAW design approach provides a way to understand how communication initiation shall be designed, but it does not provide a technical solution. A final conclusion on this matter, although not directly related to the RAW, is that we must move away from designing systems and automations where the human operator is supposed to be the backup for when the automation fails. The human may handle unexpected issues at his or her own level of LACC, but should not be forced to take over the automation's tasks. Instead, multiple layers of system support for graceful

degradation should be built in, enabling the human to continue to work on her LACC, or as a minimum, give the human time and relevant information to be able to work at and solve problems at a different LACC.

Situations where things do not work as planned (relating to **SRQ1**) were addressed in Paper V (Section 5.5), where differences and similarities in control processes in three different domains were analyzed. A common joint control issue in the domains of ATC, VTS, and TTM was identified: the shift between an overall monitoring process to particular processes-in-focus, which may result in delayed control in case of process deviations. Again, it highlights the importance of taking temporal aspects in account when designing the automation's collaboration capabilities. Flexible automation that can take all processes and the operator's situation into account is one possible solution to tackle these problems (relating to **SRQ2**).

7.2 Visualization Concepts

Two different interactive visualization concepts were developed to support the human-automation collaboration in line with the F AUTO concept, thus addressing **SRQ3**. The first visualization was a new concept for attention support, *Soft Visual Cues* (SVC), see Sections 5.1 and 5.3. It bridges the gap when the automation can identify attention, but still has not evolved into a full F AUTO automation with a complete understanding of both the operator's status and the external, controlled, processes. Exactly what a complete understanding means will differ depending on systems and processes involved. However, there is a clear difference between identifying where the operator's attention is, and interpreting what processes he or she is working with and if it is convenient to initiate communication. By providing less intrusive visual cues, the SVC complements other attention support concepts which, in contrast to a SVC, actively draw attention to a certain object or area of interest. By doing so, it enables a flexible and smooth way of supporting attention when it is possible for a system to detect that information has probably been missed, but where the automation does not have the ability to interpret the situation and understand why the information has not been seen. Thus, it is a contribution that could be applied to existing systems given that the detection of missed information is possible to achieve, and it fills a gap that has not been covered by other attention support concepts.

The second visualization concept, the glyphs, see Section 5.4, showed how the RAW principles can be used when developing a visualization for initiation of communication from an automation to a human operator, thus addressing **SRQ3**. The design was developed for initiation of communication from a future autonomous, highly automated system to a human operator using glyphs. The visualization concept was evaluated in the three domains of ATC, VTS, and TTM, and thus also addressed the cross-domain perspective in F AUTO. While being a more conceptual visualization, it provided an example of how the RAW principles can be turned into a visualization. It also showed that the basic principles can be used in different domains, especially for visualizing temporal aspects, even if the types of information to be communicated differ and must be adapted to suit the respective domain.

7.3 Human-Automation Collaboration in Context

From the results of the transversal elements of the studies, it can be concluded that principles, such as frameworks and design approaches, are needed, but when designing new automations, they must be combined with knowledge about the domain in which the automation shall operate. The need for this domain knowledge and differentiation in designs was, for example, identified in the work with the Soft Visual Cues. How soft the Soft Visual Cue should be, and what type of cues was best suited, differed between the domains depending on the need and the existing visual environment. Therefore, frameworks, design approaches, and specific visualizations must always be related to, and understood from, the context in which they shall be applied, even though they may be valid on a conceptual level. This applies to both visualization concepts as well as the RAW designed approach presented in this dissertation.

Even if there are differences between domains, there are also similarities, as shown in Paper V (Section 5.5). The importance of taking temporal aspects of human-automation collaboration in real-time control processes into account to avoid classical automation issues when introducing higher levels of automation is one example. Further research is needed, but so far there is no reason to believe other than that these commonalities could be valid in other, similar domains.

In the introduction, it was stated that automation by the F AUTO paradigm is a way of not only replacing humans, but making humans perform better together with automation. The work performed has strengthened my belief in this, and also, that regardless of level of automation, humans will always interact with the automation at some level. Hence, there will always be a need to consider the aspect of human-automation collaboration, as it will just appear at different levels. And as a final remark that cannot be repeated too many times — the key is that humans and automations shall work *together*. As soon as we start to see the human as a backup to the automation, even if considered a very skilled back-up, we have started to stray away from the cooperative view. The automation must have its own backup and even when things go wrong, the human-automation team shall solve the situation together, making the most of their respective strengths!

7.4 Future Work

My work is a small contribution to a vast research area, and a lot of research remains to be done within the area of flexible automation and human-automation collaboration. The paragraphs below summarize some of the possible ways forward that I have identified.

The first thing I would advocate is to apply the Reduced Autonomy Workspace (RAW) design approach in other contexts to explore, and refine, the design approach. This should also include the supporting glyph designs. Likewise, the Soft Visual Cues (SVC) visualization concept should be implemented and evaluated in systems from other domains than Air Traffic Control to further explore the concept. A first step in this direction is already planned, where the RAW design principles, including elements from the glyph study, are to be used in an application for Unmanned Aircraft Systems Traffic Management (UTM).

The RAW was developed as a theoretical concept with one operator and one automation. I see no reason to believe other than it can be further developed and elaborated on. To start with, I would encourage anyone to simply start to use it, hopefully to find inspiration from it, and to find potential areas for development. Already now, there are some interesting tracks to follow. One area to explore is how to use it when modelling for multiple actors and processes. It would be interesting to expand the design approach and

explore how it should look like when it includes more actors, be it human and multiple automations, multiple humans and one automation, or multiple humans and multiple automations. Another interesting track is to look at more complex situations where there are not only more actors involved, but also entangled processes that affect each other in complex ways. How to develop dynamic adaptation of information with respect to Levels of Autonomy in Cognitive Control (LACC) in communication from automation to human operators is another important question that should be addressed.

Analyzing control processes to understand a current system is a tedious work. Only relatively small amounts of data can be handled manually; longer and multiple processes are nearly impossible to analyze manually. Therefore, research in how tool support for larger analyses can be developed is needed. This could also be a first step of making such analyses in real-time to enhance flexibility of the automation, or even to use it for design of complex systems.

To achieve successful flexible automation, improved technology for input about the human operators is needed. This could include, but not be limited to, more robust eye tracking technologies and indirect measurements like real-time analysis of system interaction. A key feature that must be included in this future research is the development of methods that are truly unobtrusive. To gain operational acceptance for use of such technologies, issues regarding privacy must be addressed, as it will include handling of a lot of sensitive data. A related topic is the interpretation of the human behavior. So far, we have worked with straightforward relations between sensor data and metrics, such as attention, that are relatively easy to measure and understand. However, human behavior is very complex, and more research on how to interpret the sensor data is needed. The rapid development in AI could be a key enabler for this research. The AI development is also interesting in a wider perspective — how will it affect future automation on a more general level?

Related to gaining a better understanding of the human behavior in interaction with highly automated systems is the question about the *role* of the human in the joint cognitive system. There are several approaches for how automation shall be increased and what shall be automated, but there is a need for more research on the human role. In the work with RAW, the starting point was that humans were assumed to work with high LACC tasks

while the automation should mainly perform tasks on lower LACC, while also making sure that communication with the human is carried out on a LACC adapted to the human's situation. Releasing the human from managing transition between different LACCs should be beneficial as it is a cognitively demanding activity. However, defining these high LACC task-based roles of the humans turned out to be difficult. This is an area open for more research in order to achieve full F AUTO functionality. It would also be interesting to explore if there are situations where the automation should instead manage high LACC task while humans perform lower LACC tasks, and what implications this would have on the collaboration. Like for other areas, the development of more sophisticated AI is interesting in this respect as well — if future automations will have a completely different level of understanding about the high LACC tasks than imaginable today, the roles possible for the automation may change drastically over time, which may in turn affect the roles of the humans.

Finally, my research has targeted automation in control room environments, specifically in Air Traffic Control, but as stated in the beginning of the dissertation, the increase in automation (and digitalization) is a trend that affects many domains and sectors of the society. Hence, many of the ideas and thoughts presented in my dissertation, including the aforementioned suggestions for future work, can hopefully inspire to the development of better automation in many areas — current as well as new ones.

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